

Final Year Project



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Abstract

MedSenseAI is a web-based, AI-powered smart pharmacy platform designed to enhance medication safety in online pharmacy systems. The platform analyzes user behavior, prescriptions, and shopping cart data to detect drug–drug interactions, duplicate therapies, potential side effects, and adherence risks, and generates real-time safety and refill alerts for users. It also includes an AI chatbot that provides medication guidance and escalates high-risk queries to human pharmacists for ethical, non-diagnostic support. By combining automated safety intelligence with continuous user monitoring, the proposed system serves as a practical and scalable solution for improving medication safety and user experience in online pharmacies.

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1. Introduction

MedSenseAI is an AI-powered healthcare support platform that assists users and pharmacists in making safer medication decisions by integrating artificial intelligence with pharmacy workflows. The system enables users to upload prescriptions, check medication safety, receive AI-based recommendations, and interact with a virtual chatbot for medication guidance. It also detects potential drug interactions, analyzes prescriptions, and suggests safer medication alternatives. In addition, the platform provides pharmacists with sales analytics, customer insights, and inventory intelligence to support better decision-making. Overall, MedSenseAI aims to improve medication safety, reduce drug interaction risks, and enhance pharmacy operations through intelligent automation and data-driven insights.

2. Vision Document

2.1. Problem Statement

Table 1: Problem Statement

Category	Description
Problem	Patients often take multiple medications that may interact with each other, causing harmful side effects. Pharmacists mainly rely on manual checks or disconnected systems, which increases the risk of missing dangerous drug interactions.
Affects	Patients who rely on safe medication usage and pharmacists who must manually verify drug compatibility. Pharmacies also lack tools to analyze customer purchasing behaviour.
Impact	Drug interactions may go undetected, patients receive limited medication guidance, and pharmacies cannot effectively use data to improve sales and operations.
Solution	MedSenseAI provides an AI-driven platform that detects drug interactions, assists pharmacists in medication guidance, and analyzes pharmacy sales data to generate business insights.

2.2. Business Opportunity

The healthcare and pharmacy sector increasingly requires digital tools to improve patient safety and operational efficiency. MedSenseAI addresses this need by reducing medication errors through automated drug interaction detection, providing AI-based medical guidance for common medication queries, and enabling pharmacies to analyze customer behaviour to optimize sales strategies. Additionally, it enhances patient engagement through intelligent chatbot assistance. By integrating AI-powered healthcare support with pharmacy analytics, MedSenseAI delivers significant value to both patients and pharmacy businesses.

2.3. Objectives

The main objectives of MedSenseAI are:

1. **Improve medication safety**
Detect potential drug interactions and warn users before medication consumption.
2. **Provide AI-based medical guidance**
Allow users to interact with an AI chatbot for medication information and advice.
3. **Automate prescription analysis**
Use OCR technology to extract medicine names from uploaded prescriptions.
4. **Support pharmacists with analytics**
Provide dashboards showing sales trends, customer insights, and product demand.
5. **Enable personalized recommendations**
Suggest alternative medicines when interactions are detected.
6. **Improve pharmacy operations**
Provide lead scoring and sales optimization tools for pharmacies.

2.4. Scope

The scope of MedSenseAI includes the development of a web-based platform that supports medication safety analysis and pharmacy intelligence.

Primary features include:

- Prescription image upload and OCR processing
- Drug interaction detection.
- AI chatbot for medication guidance
- Medicine recommendation system
- Refill reminders.
- Pharmacist dashboard with analytics
- Lead scoring and sales optimization.

The system will be accessible through a web interface and will support both users and pharmacists.

2.5. Constraints

The following constraints may affect the project:

- Access to real pharmaceutical datasets may be limited.
- AI models may require synthetic or publicly available data for training.
- The system must comply with data privacy and healthcare regulations.
- Development must be completed within the academic project timeline.
- System performance must support multiple concurrent users.

2.6. Stakeholder and User Descriptions

The stakeholder profiles and user-related attributes of the MedSenseAI system are described in this section.

2.6.1. Market Demographics

The system targets:

- Patients who require medication safety guidance
- Pharmacists managing prescriptions and medicine sales
- Pharmacy owners seeking analytics insights
- Healthcare organizations interested in AI-based medication support

2.6.2. Stakeholder Summary

Table 2: Stakeholder Summary

Stakeholders	Description	Responsibilities
Patients / General Users	Individuals who use the system to upload prescriptions, check drug interactions, and receive AI-based medication guidance.	Upload prescriptions, check medication safety alerts.
Pharmacists	Healthcare professionals who review prescriptions and provide medication guidance when necessary.	Monitor prescription analysis results, verify drug interactions, and assist patients with advice.
Pharmacy Owners / Managers	Decision-makers responsible for pharmacy operations and business insights.	Monitor sales analytics, review medicine demand trends, and optimize pharmacy inventory and sales strategies.
Developers	Technical stakeholders responsible for system development.	Design and develop system features, integrate AI modules, maintain the database, and ensure system reliability.
Supervisors / Panel	Academic reviewers and evaluators	Review project documentation, provide feedback, and evaluate the final system and research work.

2.6.3. User Environment

MedSenseAI is a web-based platform accessible through any modern browser on desktop computers, tablets, or mobile devices with a stable internet connection. The system provides an intuitive user interface that allows patients and pharmacists to easily access medication information, check potential drug interactions, and receive AI-assisted guidance. Pharmacies can also utilize the platform to analyse customer purchasing behaviour and gain insights for improving sales strategies. The backend infrastructure operates on cloud-based services that support secure database storage, AI processing, and API integration to ensure efficient system performance and reliable data management.

2.6.4. Stakeholder Profiles

2.6.4.1. *Project Supervisor*

Table 3: Supervisor of the Project

Representatives	Supervisor: Mr. Umer Iqbal Co-Supervisor: Dr. Usama
Description	The supervisors are responsible for guiding and monitoring the project development process, ensuring that all activities align with the project goals and academic requirements.
Type	Technical stakeholders with expertise in the project domain.
Responsibilities	1. Provide technical guidance to the development team. 2. Ensure the project is completed within the defined timeline. 3. Monitor system development and ensure it meets required standards. 4. Review project documentation and design artifacts. 5. Assist in resolving technical challenges during development.
Success Criteria	Successful completion and delivery of a functional system that meets the defined requirements and project objectives.
Involvement	1. Requirement reviewer 2. Progress evaluator 3. Technical advisor
Comments	None

2.6.4.2. *Development Team of the Project*

Table 4: Development Team of the Project

Representatives	• Iman Fatima • Amna Rana • Shawal Ahmad
Description	The team responsible for the design, development, testing, and deployment of MedSenseAI.
Type	Technical stakeholders skilled in software engineering, program analysis, and visualization.
Responsibilities	1. Implement AI-based drug interaction detection and medication guidance features. 2. Perform testing to ensure accuracy and efficiency. 3. Maintain project documentation and version control. 4. Ensure scalability and user-friendliness of the system.
Success Criteria	Complete and stable implementation of all planned system modules with verified testing.
Involvement	1. Developers 2. Testers 3. UI/UX designers
Deliverables	1. System documentation 2. Fully functional web application
Comments	none

3. System Requirements Specification

3.1. System Features

MedSenseAI provides the following features:

- User authentication and role management
- Prescription upload and OCR processing
- Drug interaction detection.
- AI chatbot medication support
- Medicine recommendation engine
- Medication refill alerts
- Pharmacist dashboard
- Sales analytics and reporting
- Lead scoring and sales optimization.
- Notifications and alerts
- Data visualization dashboards

3.2. Functional Requirements

The MedSenseAI system shall fulfill the following functional requirements to ensure effective medication safety, AI-assisted guidance, and pharmacy analytics for improved patient care and pharmacy operations.

3.2.1. Authentication and User Management

The system shall allow users to register and log in securely.

The system shall support role-based access control for:

- Users
- Pharmacists
- Administrators

3.2.2. Prescription Upload

- The system shall allow users to upload prescription images.
- The system shall extract medicine names using OCR.
- The extracted data shall be stored in the system database.

3.2.3. Drug Interaction Detection

- The system shall analyze medicines from prescriptions
- The system shall detect potential interactions between medications.
- The system shall display warnings if interactions are detected.

3.2.4. Medicine Recommendation

- The system shall suggest safer alternatives, if harmful interactions are detected.
- Recommendations shall be generated using AI-based analysis.

3.2.5. AI Chatbot Assistance

- The system shall allow users to ask medication-related questions.
- The chatbot shall provide predefined responses for common queries.
- The chatbot may escalate the query to a pharmacist, if the question cannot be answered.

3.2.6. Refill Reminder System

- The system shall track medication usage.
- The system shall notify users when medication refills are due.

3.2.7. Pharmacist Dashboard

Pharmacists shall access a dashboard displaying:

- Prescription analysis
- Interaction alerts
- Customer insights
- Sales statistics

3.2.8. Lead Scoring

- The system shall analyze customer purchasing patterns.
- Customers shall be assigned a lead score based on purchase behaviour.
- This score helps pharmacies identify potential repeat customers.

3.2.9. Sales Optimization

- The system shall analyze pharmacy sales data.
- It shall provide recommendations for product promotion and stock optimization.

3.2.10. Reports and Analytics

The system shall generate visual reports including:

- Sales trends
- Customer behaviour
- Medication Demand

Reports shall be exportable.

3.3. Non-Functional Requirements

The MedSenseAI system shall adhere to the following non-functional requirements to ensure reliability, performance, and usability across diverse environments.

3.3.1. Performance Requirements

- The system shall analyze prescriptions and detect potential drug interactions within **5 seconds** for standard queries.
- The system shall efficiently support multiple concurrent users including patients and pharmacists without noticeable performance degradation.
- The system shall optimize AI processing and database queries to ensure quick loading of dashboards, chatbot responses, and medication analysis.

3.3.2. Usability Requirements

- The system shall provide a **user-friendly graphical interface** with intuitive controls for navigation and visualization.
- The system shall offer tooltips, legends, and labels to assist users in interpreting CFG elements.
- The interface shall maintain consistency in layout, colors, and symbols for better comprehension.

3.3.3. Reliability Requirements

- The system shall ensure that all generated CFGs and coverage reports are **accurate and verifiable**.
- The system shall recover from minor failures such as temporary server issues or interrupted requests without affecting stored data.
- The platform shall maintain data consistency during operations such as prescription uploads, interaction analysis, and sales data processing.

3.3.4. Security Requirements

- The system shall require secure user authentication before accessing system functionalities.
- The system shall store sensitive user and medical data using secure encryption standards.
- The system shall implement role-based access control to prevent unauthorized access to patient information and pharmacy data.

3.3.5. Compatibility Requirements

- The system shall be compatible with major web browsers (Chrome, Edge, Firefox).
- The application shall function properly on desktop computers, tablets, and smartphones.
- The platform shall maintain consistent performance and responsive design across different devices and operating systems.

3.3.6. Maintainability Requirements

- The system shall follow a modular and well-documented architecture to simplify maintenance and upgrades.
- Developers shall be able to extend system features such as adding new AI models or analytics modules without affecting existing components.
- The system shall maintain proper documentation and version control for continuous development.

3.3.7. Portability Requirements

- The system shall be deployable on cloud-based infrastructure to support scalable healthcare services.
- The application shall be accessible through web browsers without requiring additional software installation.

3.3.8. Scalability Requirements

- The system shall support a growing number of users, prescriptions, and pharmacy transactions without reducing system performance.
- The architecture shall allow expansion of AI services, databases, and servers as system demand increases.

4. Use Case Diagram



Figure 1: Use case Diagram

5. Expanded Use Case

5.1. User Authentication

Table 5: EUC-01 User Authentication

ID	EUC-01
Use Case	User Authentication
Scope:	MedSenseAI System
Primary Actor	User
Stakeholders And Interests:	Admin: Wants secure access control and proper user management. User: Wants secure and easy access to system services. Organization: Needs secure authentication and proper role management.
Pre-condition	• User must have internet connectivity. • User must have an account for login.
Post-condition	• User account is authenticated. • User is redirected to the system dashboard.
Main Success Scenario	1. The user selects Register or Login. 2. The system displays the required input fields. 3. The user enters valid credentials (email, password). 4. The system validates credentials and user role. 5. If valid, the user is redirected to their dashboard.
Alternate Flow	4.1(a) If credentials are incorrect → System displays error message. 4.1(b) If user does not have account → User completes registration process.

5.2. Upload Prescription

Table 6: EUC-02 Upload Prescription

ID	EUC-02
Use Case	Upload Prescription
Scope:	MedSenseAI System
Primary Actor	User
Stakeholders And Interests:	User: Wants automatic extraction of medicines from prescription. Pharmacist: Needs accurate prescription interpretation. Organization: Wants automated prescription processing.
Pre-condition	• User must be logged into the system. • User must have a prescription image.
Post-condition	• Prescription is stored in the system. • Medicines are extracted from the prescription
Main Success Scenario	1. User navigates to Upload Prescription section. 2. User selects prescription image file. 3. User uploads the prescription. 4. System receives the image. 5. System processes the image using OCR. 6. System extracts medicine names. 7. System stores extracted data. 8. System displays extracted medicines.
Alternate Flow	5.1(a) If OCR confidence is low → System asks user to confirm medicines manually.

5.3. Add Medicine to Cart

Table 7: EUC-03 Add Medicine to Car

ID	EUC-03
Use Case	Add Medicine to Cart
Scope:	MedSenseAI System
Primary Actor	User
Stakeholders And Interests:	User: Wants to purchase medicines easily. Organization: Wants smooth purchasing workflow.
Pre-condition	• User must be logged in. • Medicine must exist in inventory.
Post-condition	• Selected medicine is added to the cart.
Main Success Scenario	1. User searches for a medicine. 2. System displays available medicines. 3. User selects a medicine. 4. User clicks Add to Cart. 5. System updates cart items. 6. System confirms the addition.
Alternate Flow	3.1(a) Medicine not available → System shows Out of Stock message.

5.4. Drug Interaction Check

Table 8: EUC-04 Drug Interaction Check

ID	EUC-04
Use Case	Drug Interaction Check
Scope:	MedSenseAI System
Primary Actor	User
Stakeholders And Interests:	User: Wants safe medication usage. Healthcare providers: Want reduced medication risks
Pre-condition	• User has medicines in cart or prescription list.
Post-condition	• System displays interaction warnings if detected.
Main Success Scenario	1. User adds medicine to cart. 2. System retrieves list of medicines. 3. System checks drug interaction database. 4. System analyses medicine combinations. 5. System detects potential interactions. 6. System displays warning message.
Alternate Flow	4.1(a) If no interaction detected → System confirms medicine safety.

5.5. View Medicine Recommendations

Table 9: EUC-05 View Medicine Recommendations

ID	EUC-05
Use Case	View Medicine Recommendations
Scope:	MedSenseAI System
Primary Actor	User
Stakeholders And Interests:	User: Wants alternative medicine suggestions. Pharmacist: Wants safer medication options.
Pre-condition	User has prescription or selected medicines.
Post-condition	System displays recommended alternatives.
Main Success Scenario	1. User requests medicine recommendation. 2. System analyses prescription data. 3. System runs recommendation algorithm. 4. System generates alternative medicines. 5. System displays recommendations.
Alternate Flow	3.1(a) If no alternatives found → System notifies user.

5.6. Refill Alert System

Table 10: EUC-06 Refill Alert System

ID	EUC-06
Use Case	Refill Alert System
Scope:	MedSenseAI System
Primary Actor	User
Stakeholders And Interests:	User: Wants timely reminders for medication refill. Healthcare system: Wants improved medication adherence.
Pre-condition	User medication history exists.
Post-condition	Refill notification is sent.
Main Success Scenario	1. System monitors medication history. 2. System estimates refill schedule. 3. System detects upcoming refill date. 4. System sends reminder notification. 5. User receives alert.
Alternate Flow	3.1(a) If reminder ignored → System sends follow-up reminder.

5.7. AI Chatbot Assistance

Table 11: EUC-07 AI Chatbot Assistance

ID	EUC-07
Use Case	AI Chatbot Assistance
Scope:	MedSenseAI System
Primary Actor	Pharmacist
Stakeholders And Interests:	Pharmacist: Wants quick access to medicine information, sales trends, and inventory insights. Organization: Wants faster decision-making and efficient pharmacy operations.
Pre-condition	- Pharmacist must be logged into the system. - System database must contain medicine and sales data.
Post-condition	Pharmacist receives AI-generated insights or answers related to medicines or sales data.
Main Success Scenario	- Pharmacist opens the AI Assistant interface. - Pharmacist enters a query (e.g., medicine sales trend, stock information, drug details). - System sends the query to the AI engine. - AI processes the request using available system data. - AI generates insights or recommendations. - System displays the response to the pharmacist.
Alternate Flow	4.1(a) If requested data is unavailable → System notifies pharmacist that information cannot be retrieved.

5.8. Pharmacist Guidance for Drug Interaction

Table 12: EUC-08 Pharmacist Guidance for Drug Interaction

ID	EUC-08
Use Case	Pharmacist Guidance for Drug Interaction
Scope:	MedSenseAI System
Primary Actor	Pharmacist
Stakeholders And Interests:	User: Wants safe medication guidance. Pharmacist: Wants to ensure safe prescriptions and prevent harmful drug interactions. Organization: Wants to reduce medication risks.
Pre-condition	- User adds medicines to cart or uploads prescription. - Drug Interaction Engine detects a potential interaction.
Post-condition	Pharmacist provides guidance or recommendation to the user.
Main Success Scenario	- User uploads prescription or adds medicines to cart. - System runs Drug Interaction Engine. - System detects a potential drug interaction. - System sends alert to the pharmacist. - Pharmacist reviews the interaction details. - Pharmacist provides recommendation or guidance. - System displays pharmacist guidance to the user
Alternate Flow	3.1(a) If no interaction detected → System allows normal purchase. 4.1(b) If pharmacist unavailable → System displays warning and asks user to consult pharmacist later

5.9. View KPI Dashboard

Table 13: EUC-09 View KPI Dashboard

ID	EUC-09
Use Case	View KPI Dashboard
Scope:	MedSenseAI System
Primary Actor	Pharmacist
Stakeholders And Interests:	Pharmacist: Wants performance insights. Organization: Wants sales analytics.
Pre-condition	Pharmacist must be logged in.
Post-condition	Dashboard analytics displayed.
Main Success Scenario	- Pharmacist logs into system. - Pharmacist opens dashboard. - System retrieves sales data. - System calculates KPIs. - System displays analytics.
Alternate Flow	3.1(a) No data available → System displays empty dashboard.

5.10. Manage Inventory

Table 14: EUC-10 Manage Inventory

ID	EUC-10
Use Case	Manage Inventory
Scope:	MedSenseAI System
Primary Actor	Pharmacist
Stakeholders And Interests:	Pharmacist: Wants to update medicine stock. Organization: Wants accurate inventory records.
Pre-condition	Pharmacist authenticated.
Post-condition	Inventory is updated.
Main Success Scenario	- Pharmacist opens inventory page. - Pharmacist selects medicine. - Pharmacist updates stock quantity. - System saves changes. - System confirms update.
Alternate Flow	2.1(a) Pharmacist adds new medicine.

5.11. Manage Orders

Table 15: EUC-11 Manage Orders

ID	EUC-11
Use Case	Manage Orders
Scope:	MedSenseAI System
Primary Actor	Pharmacist
Stakeholders And Interests:	Pharmacist: Wants to manage customer orders. User: Wants order processing
Pre-condition	Orders exist in the system.
Post-condition	Order status updated.
Main Success Scenario	- Pharmacist views orders list. - Pharmacist selects an order. - Pharmacist reviews details. - Pharmacist updates order status. - System saves changes.
Alternate Flow	3.1(a) Order cancelled.

5.12. Lead Scoring

Table 16: EUC-12 Lead Scoring

ID	EUC-12
Use Case	Lead Scoring
Scope:	MedSenseAI System
Primary Actor	Pharmacist
Stakeholders And Interests:	Pharmacist: Wants to identify high-value customers and prioritize engagement. Organization: Wants better sales conversion and customer targeting.
Pre-condition	- Pharmacist must be logged into the system. - Customer interaction and purchase data must exist in the system.
Post-condition	System generates lead scores for customers based on behavior and purchase history.
Main Success Scenario	- Pharmacist opens the Lead Scoring module. - System retrieves customer interaction and purchase data. - System analyses customer behaviour using the lead scoring model. - System calculates lead scores for customers. - System ranks customers based on lead scores. - System displays prioritized customer list to the pharmacist.
Alternate Flow	3.1(a) If insufficient customer data → System displays message indicating limited scoring accuracy.

5.13. Sales Optimization

Table 17: EUC-13 Sales Optimization

ID	EUC-13
Use Case	Sales Optimization
Scope:	MedSenseAI System
Primary Actor	Pharmacist
Stakeholders And Interests:	Pharmacist: Wants insights to improve sales performance. Organization: Wants optimized product placement and better sales strategy.
Pre-condition	- Pharmacist must be logged into the system. - Sales data must be available in the database.
Post-condition	System provides optimized sales recommendations.
Main Success Scenario	- Pharmacist opens the Sales Optimization module. - System retrieves historical sales data. - System analyses product performance and sales patterns. - System identifies high-selling and low-selling medicines. - System generates sales optimization recommendations. - System displays insights and suggestions to the pharmacist.
Alternate Flow	3.1(a) If sales data unavailable → System displays message indicating insufficient data.

6. System Sequence Diagram

6.1. User Authentication

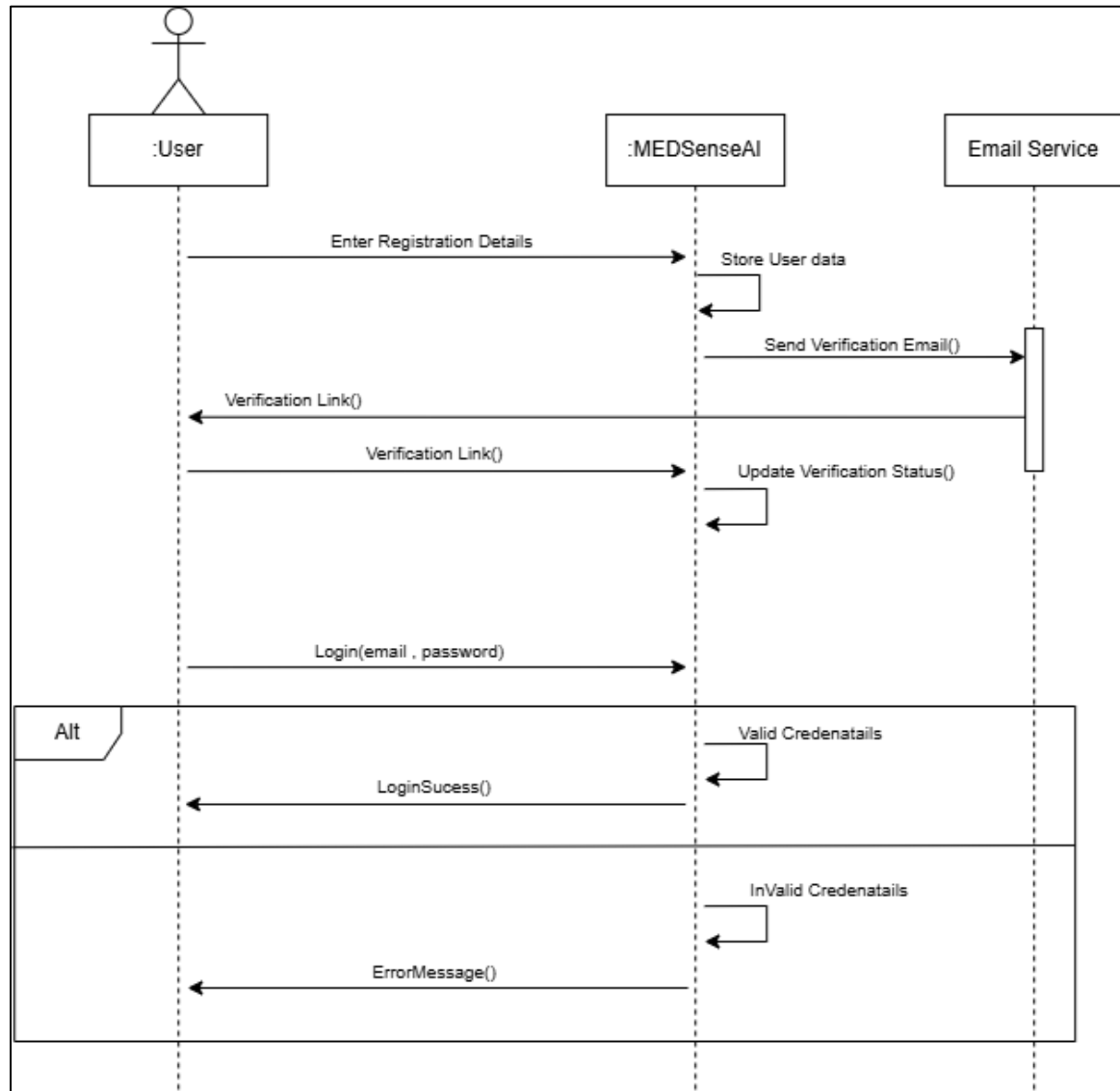


Figure 2:SSD-01 User Authentication

6.2. Upload Prescription

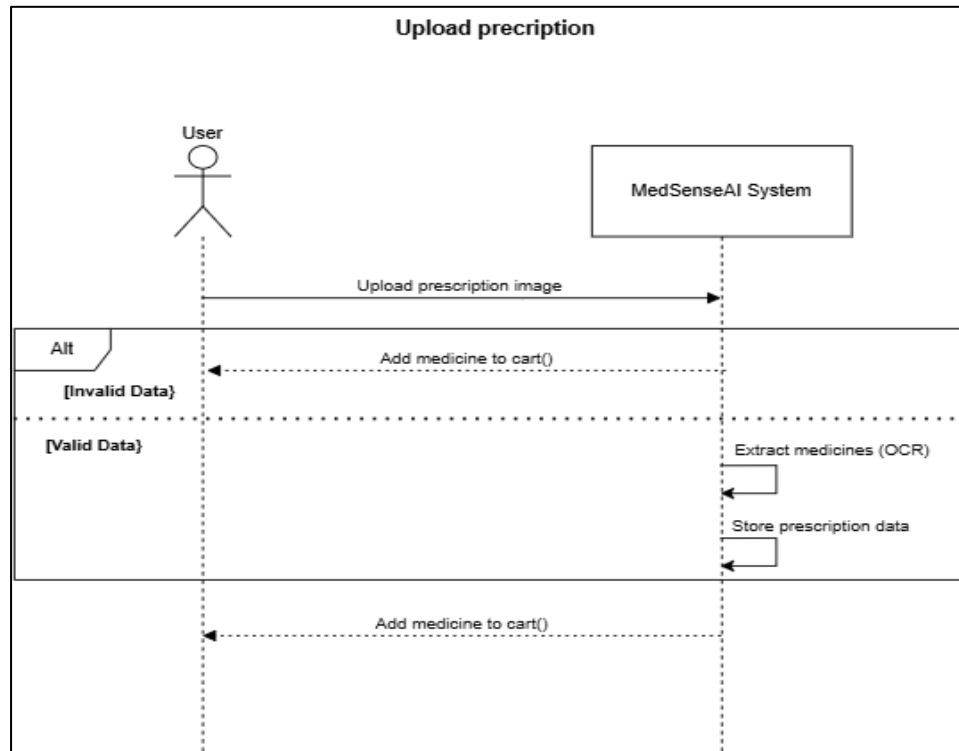


Figure 3: SSD-02 Upload Prescription

6.3. Interaction Detection

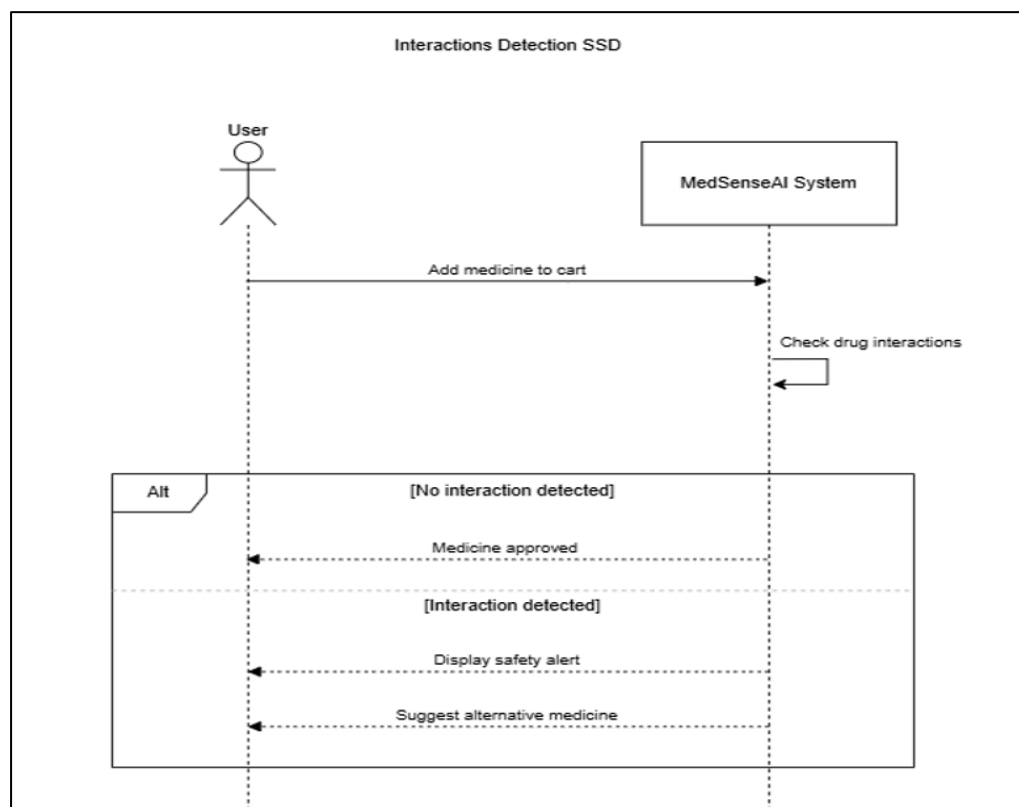


Figure 4: SSD-03 Interaction Detection

6.4. Chatbot Supplier

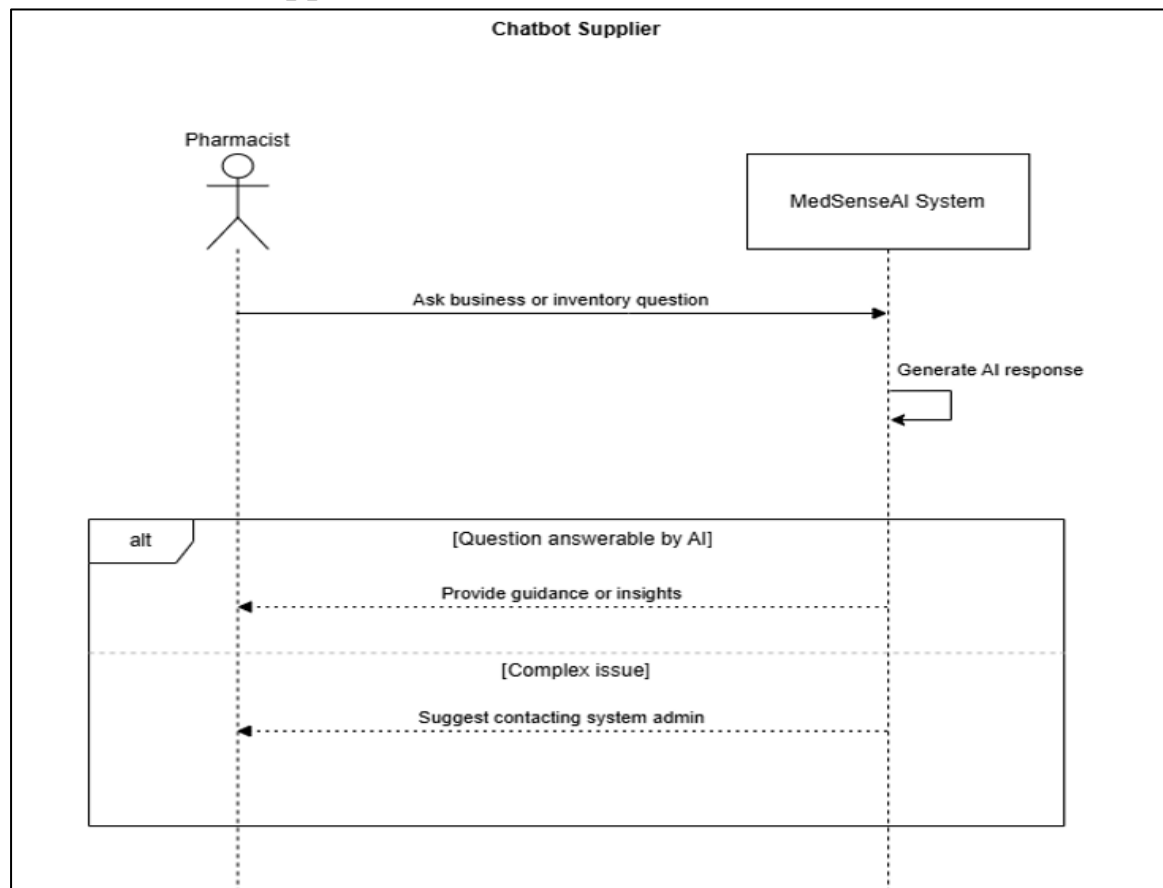


Figure 5: SSD-04 Chatbot Supplier

6.5. Escalate to Pharmacist

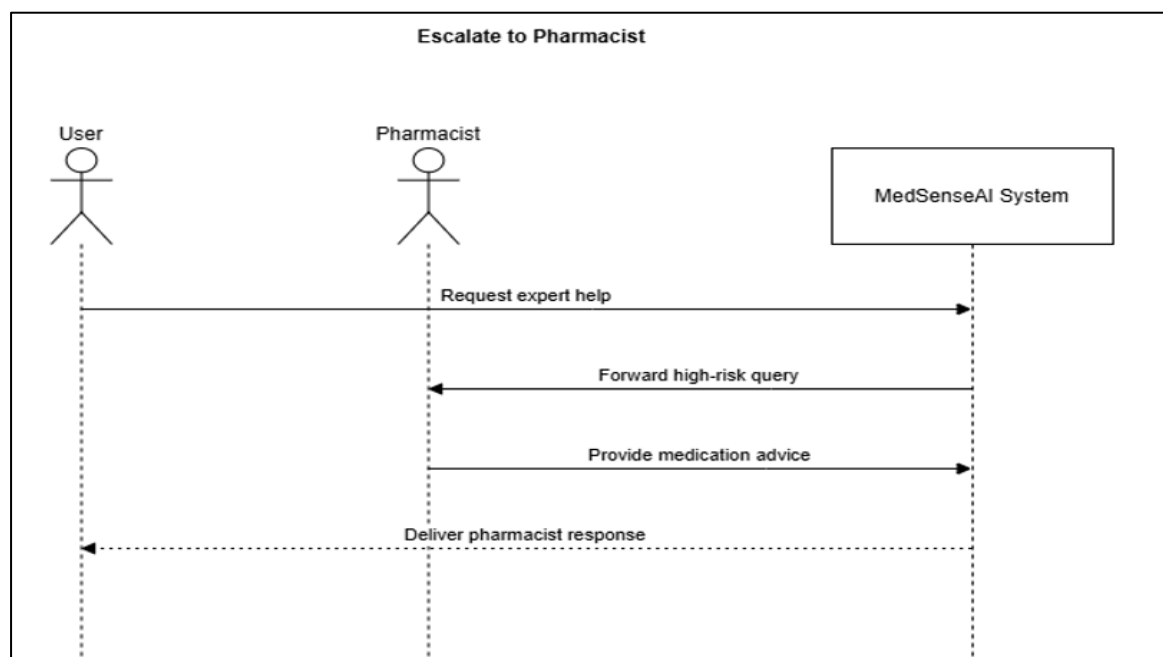


Figure 6: SSD-05 Escalate to Pharmacist

6.6. Lead Scoring Analytics

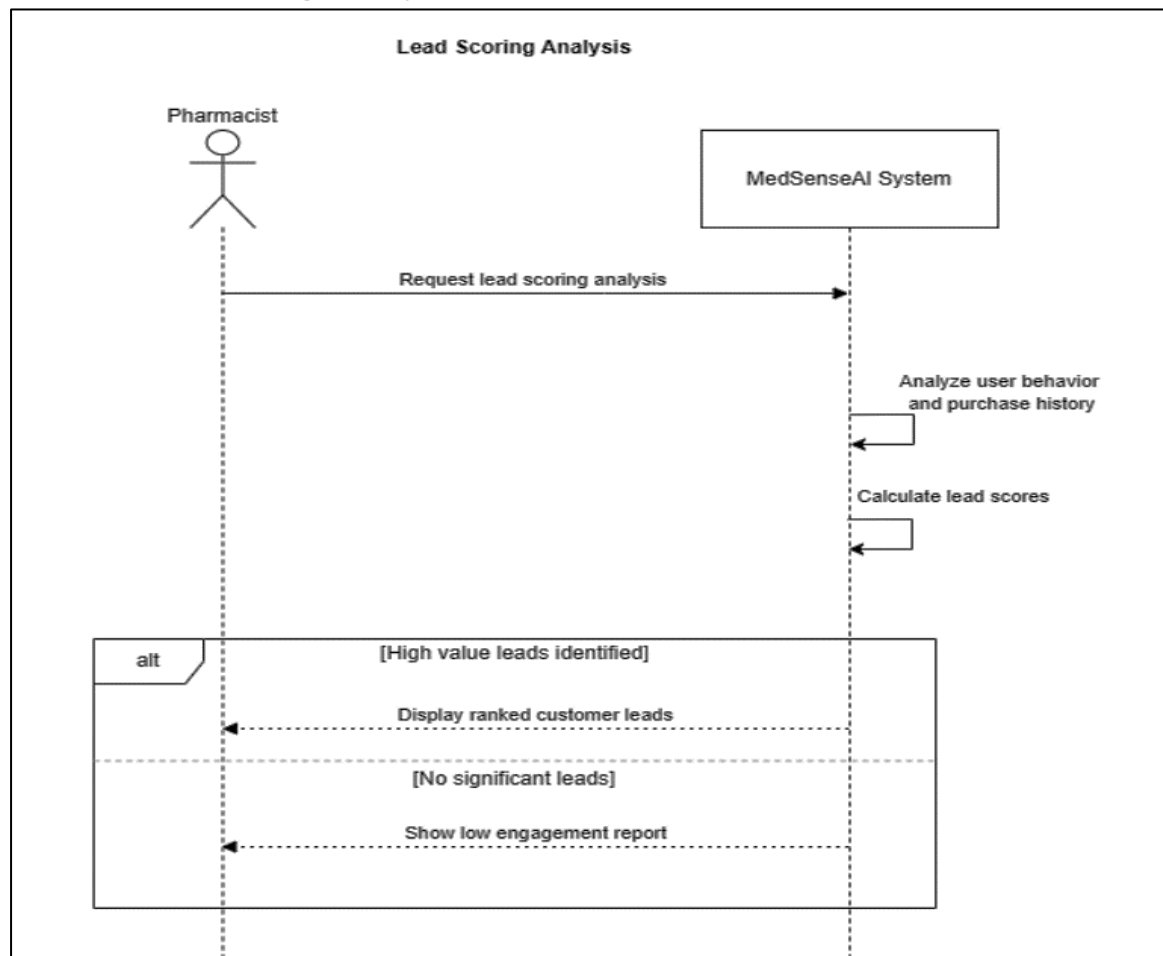


Figure 7: SSD-06 Lead Scoring Analytics

6.7. Manage Inventory

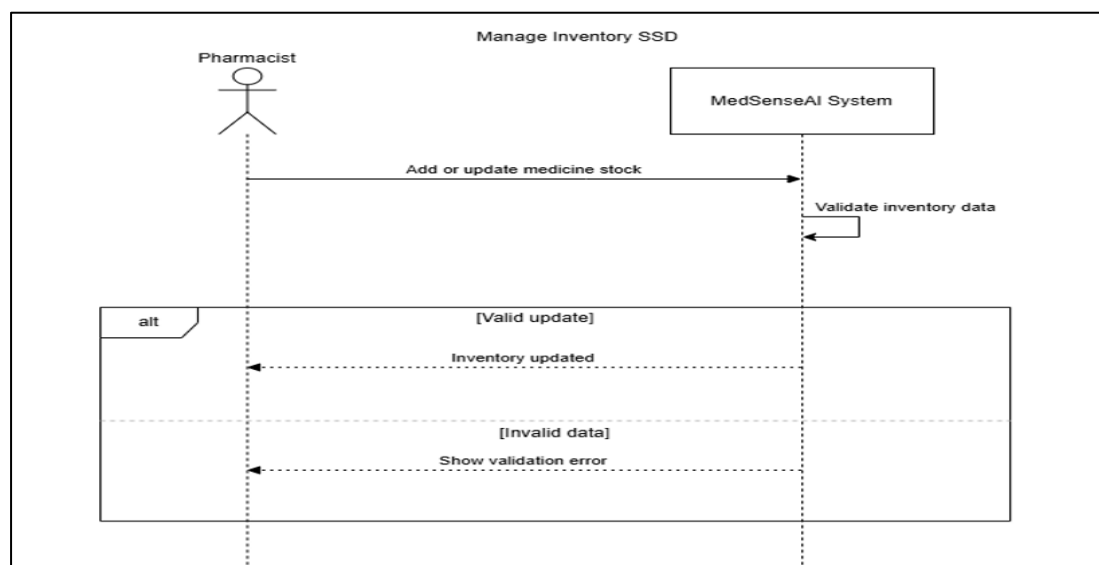


Figure 8: SSD-07 Manage Inventory

6.8. Medicine Recommendations

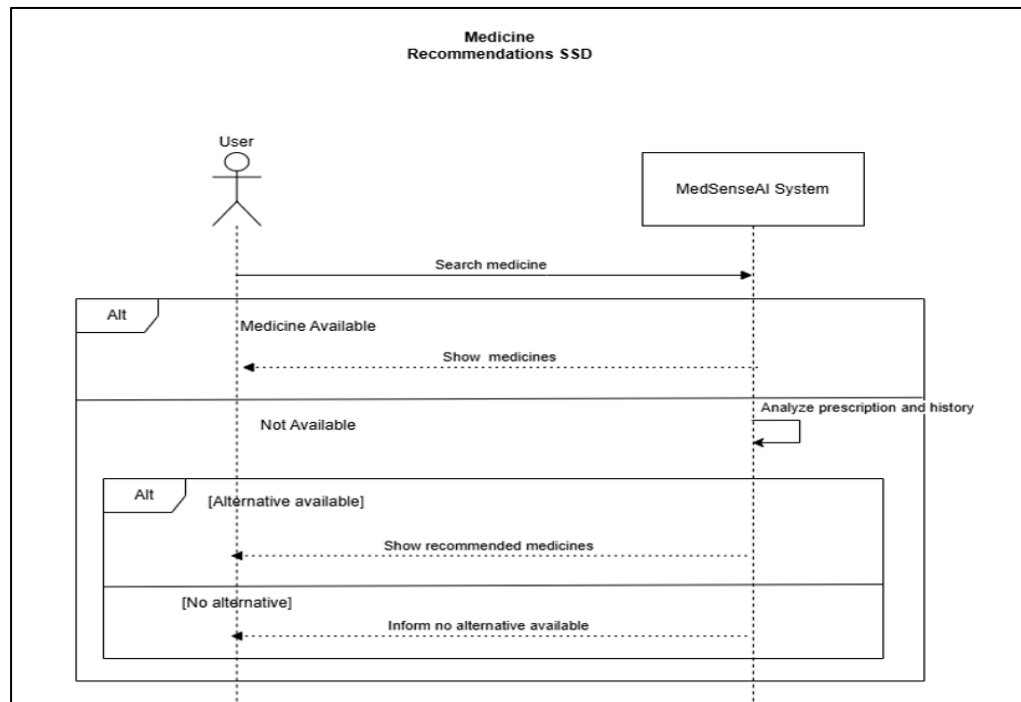


Figure 9: SSD-08 Medicine Recommendation

6.9. Refill

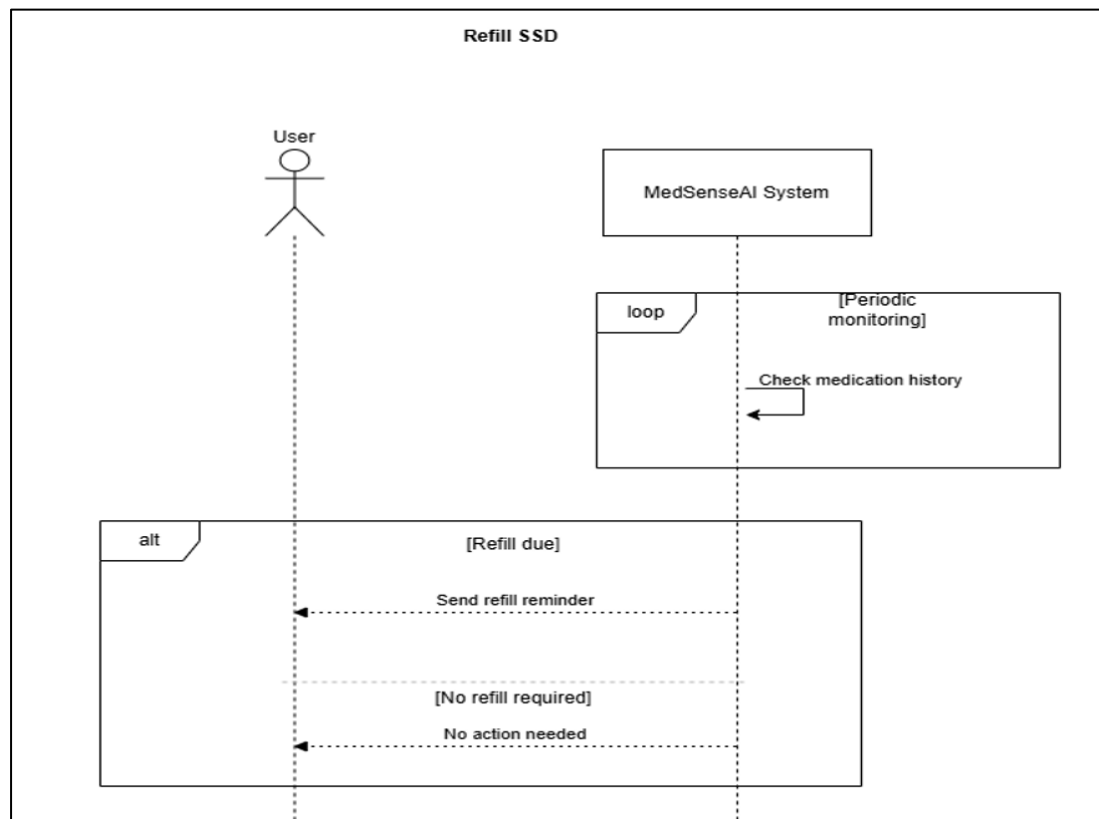


Figure 10: SSD-09 Refill

6.10. Sales Insights

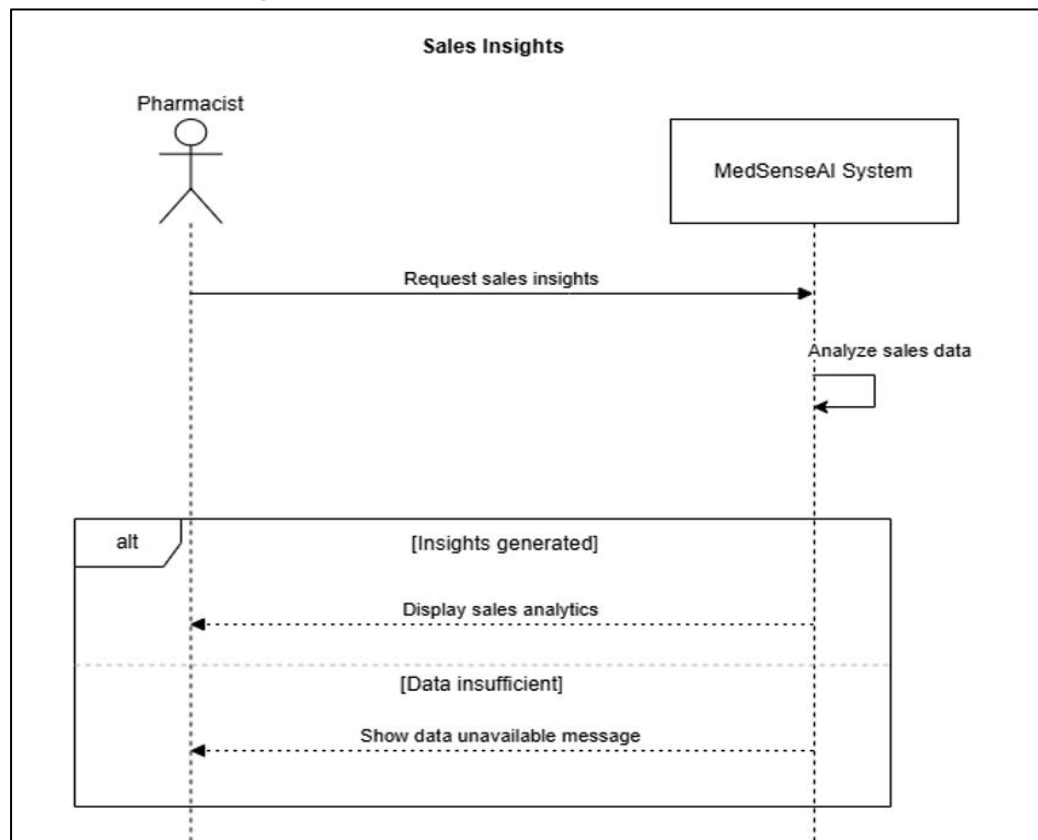


Figure 11: SSD-10 Sales Insights

6.11. Sales Optimization

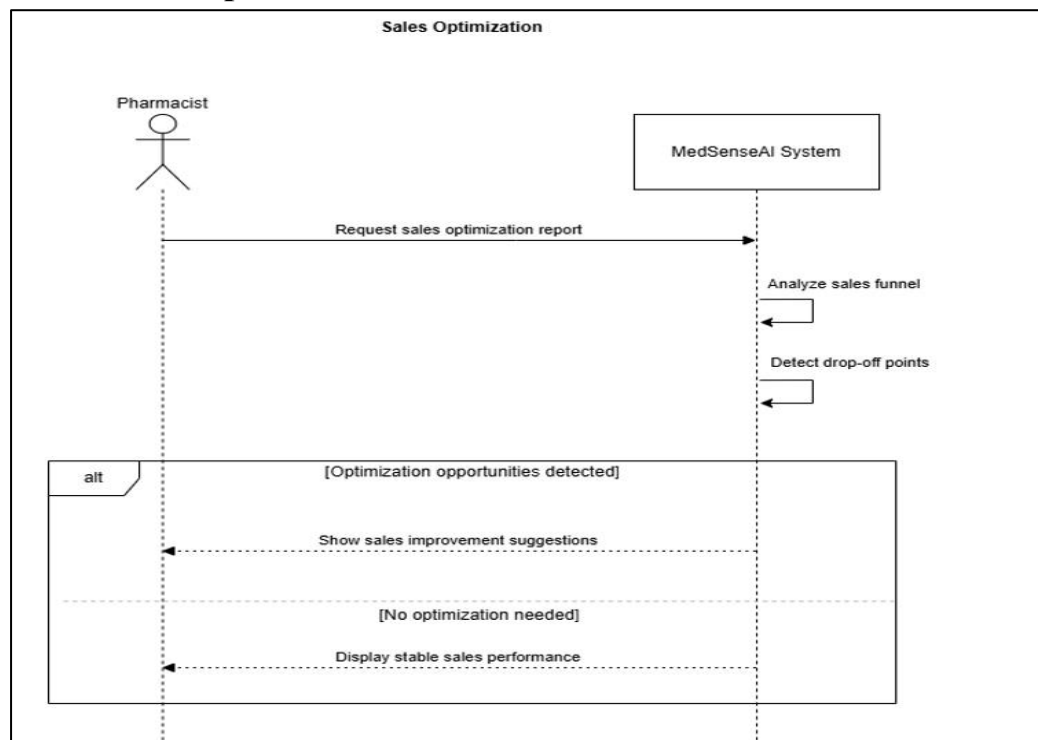


Figure 12: SSD Sales Optimization

7. Sequence Diagram

7.1. User Authentication

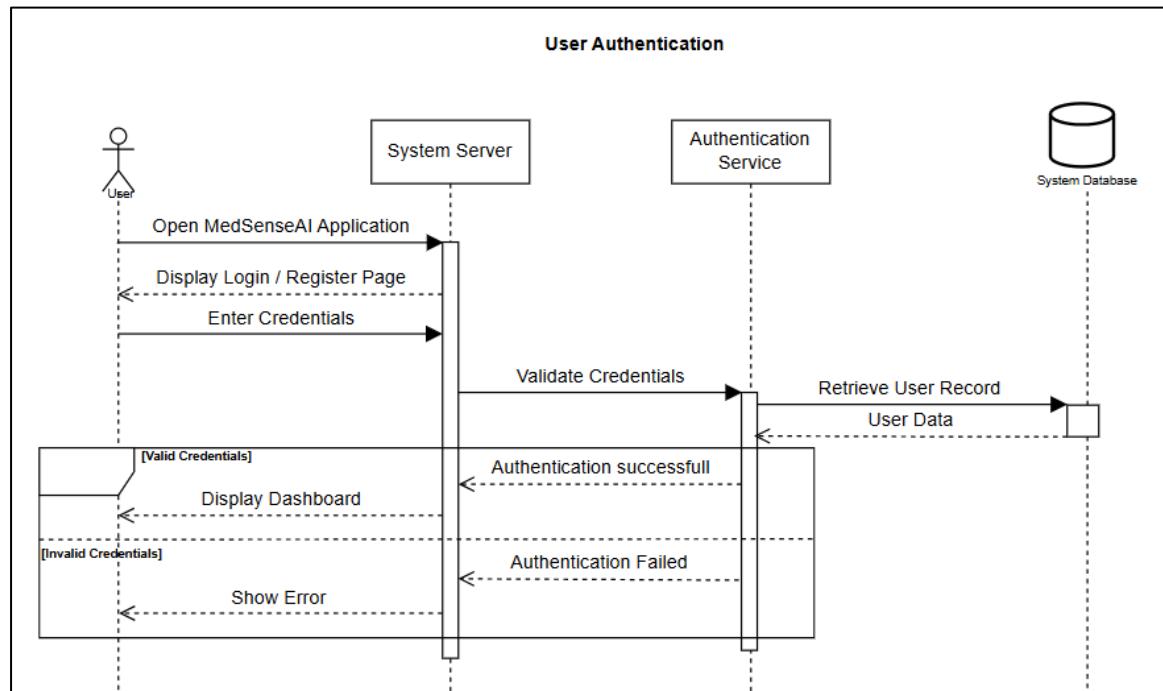


Figure 13:SD-01 User Authentication

7.2. Upload Prescription

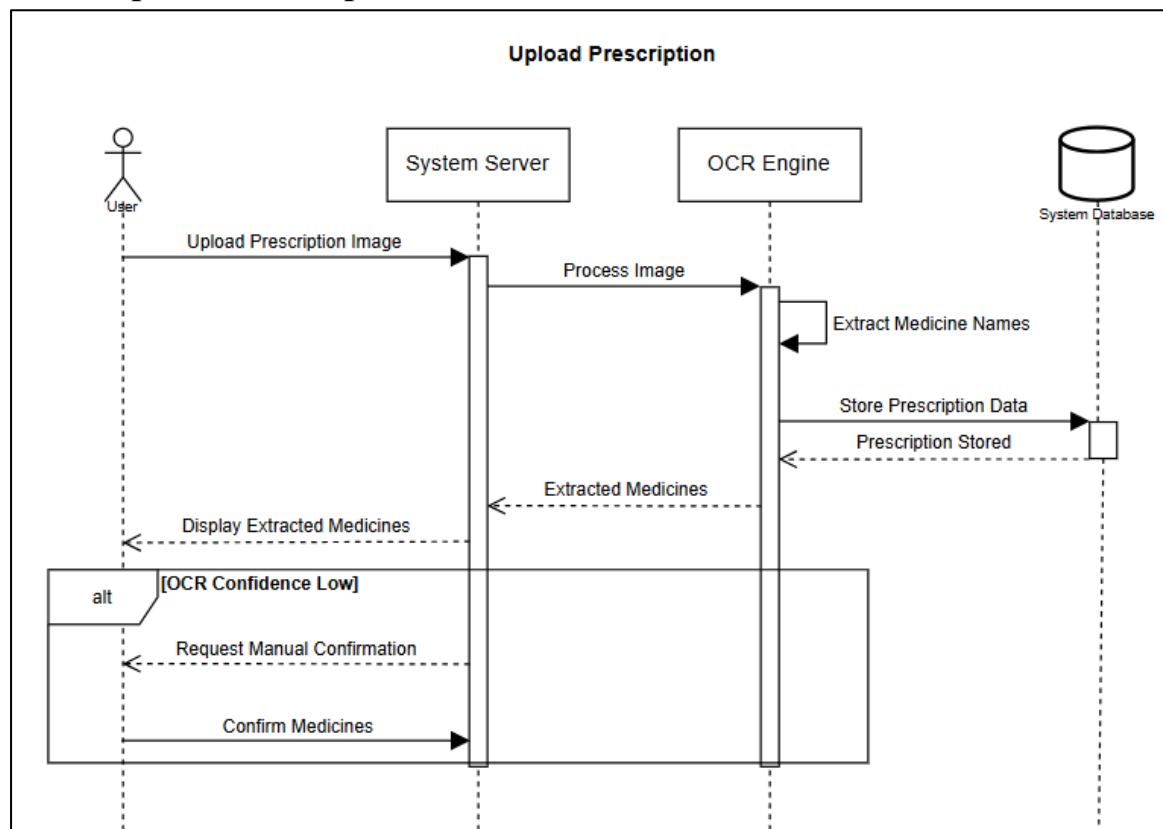


Figure 14: SD-02 Upload Prescription

7.3. Drug Interaction

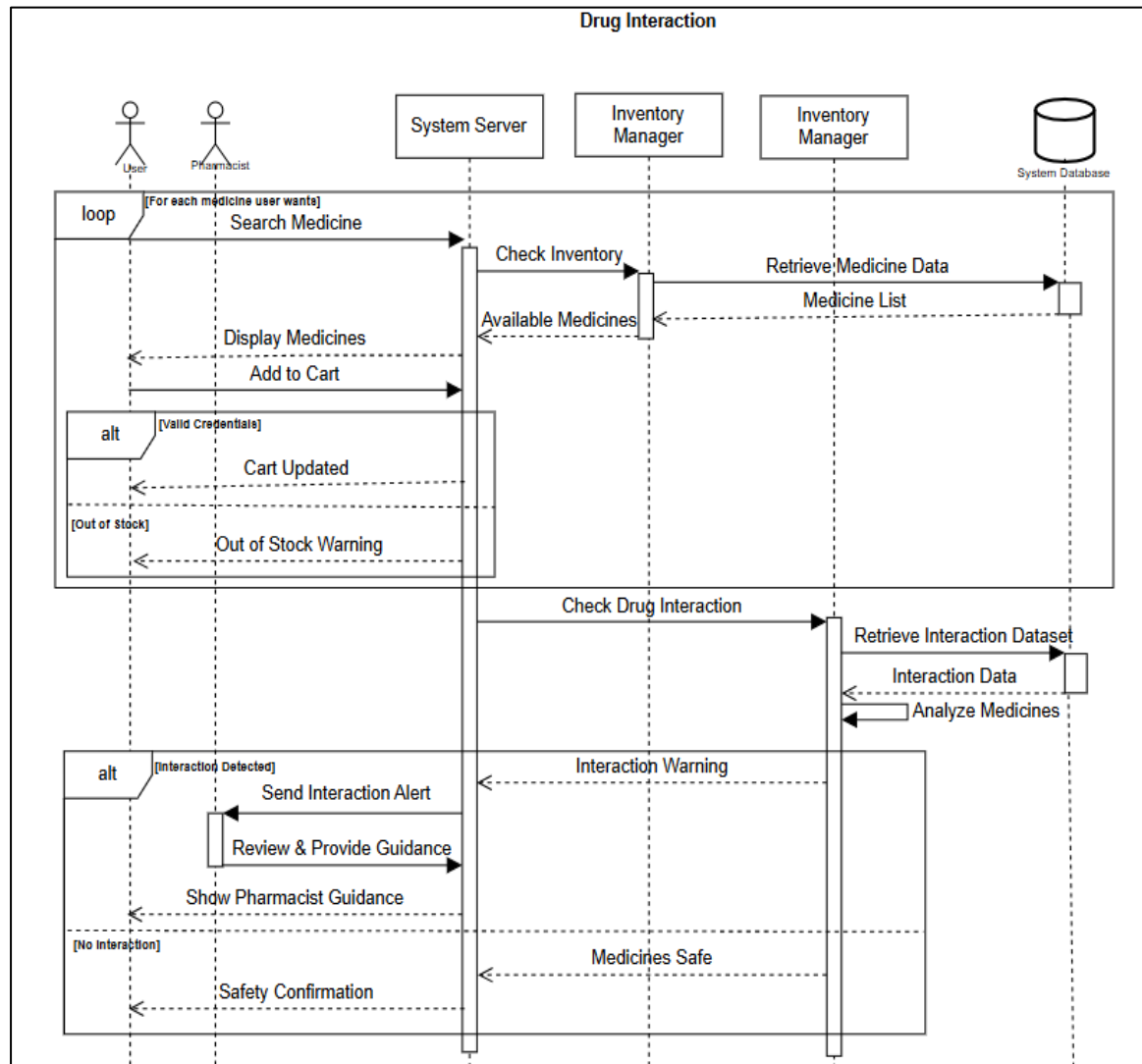


Figure 15: SD-03 Drug Interaction

7.4. Order Placement and Refill Notification

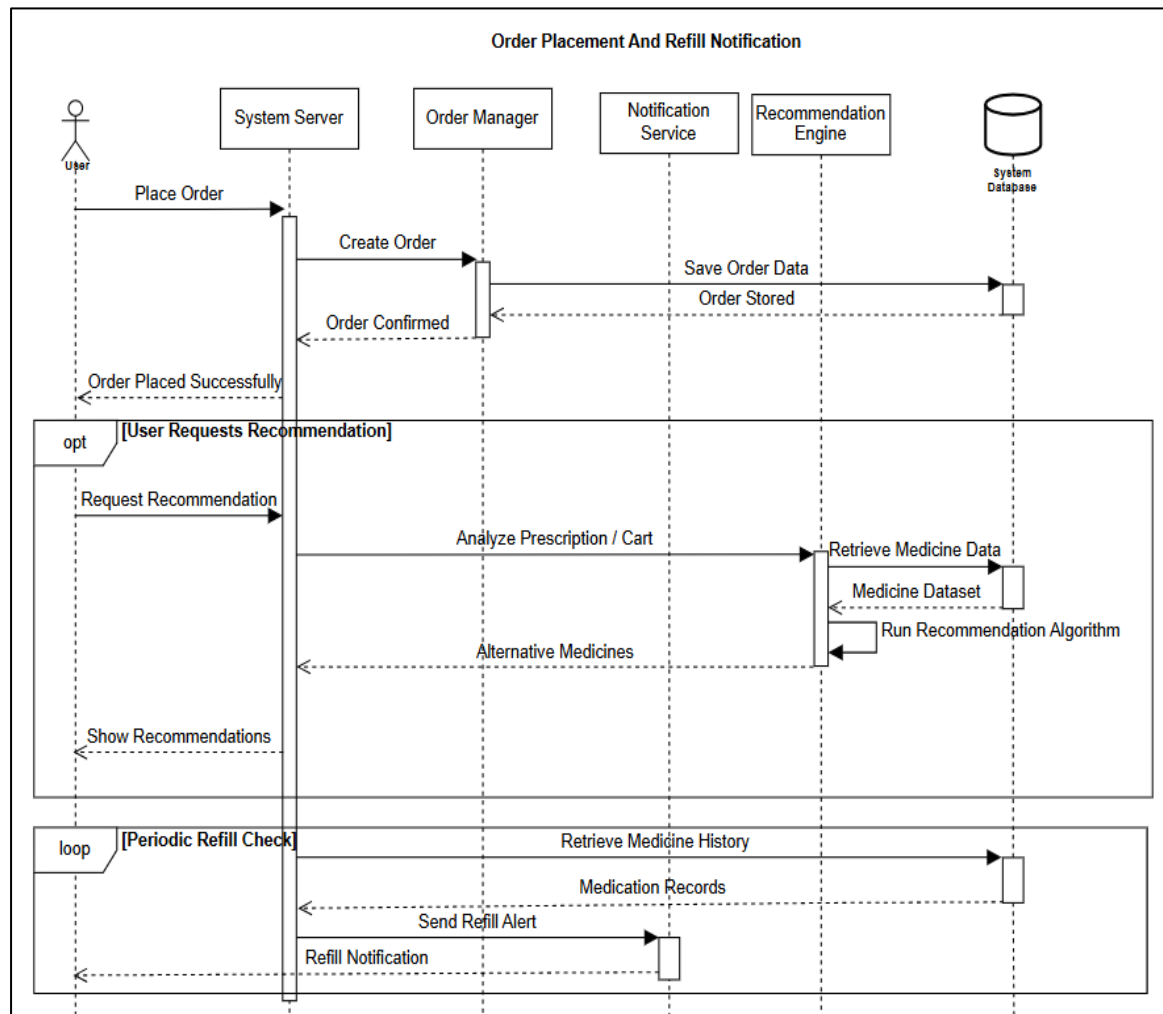


Figure 16: SD-04 Order Placement and Refill Notification

7.5. Pharmacist Authentication

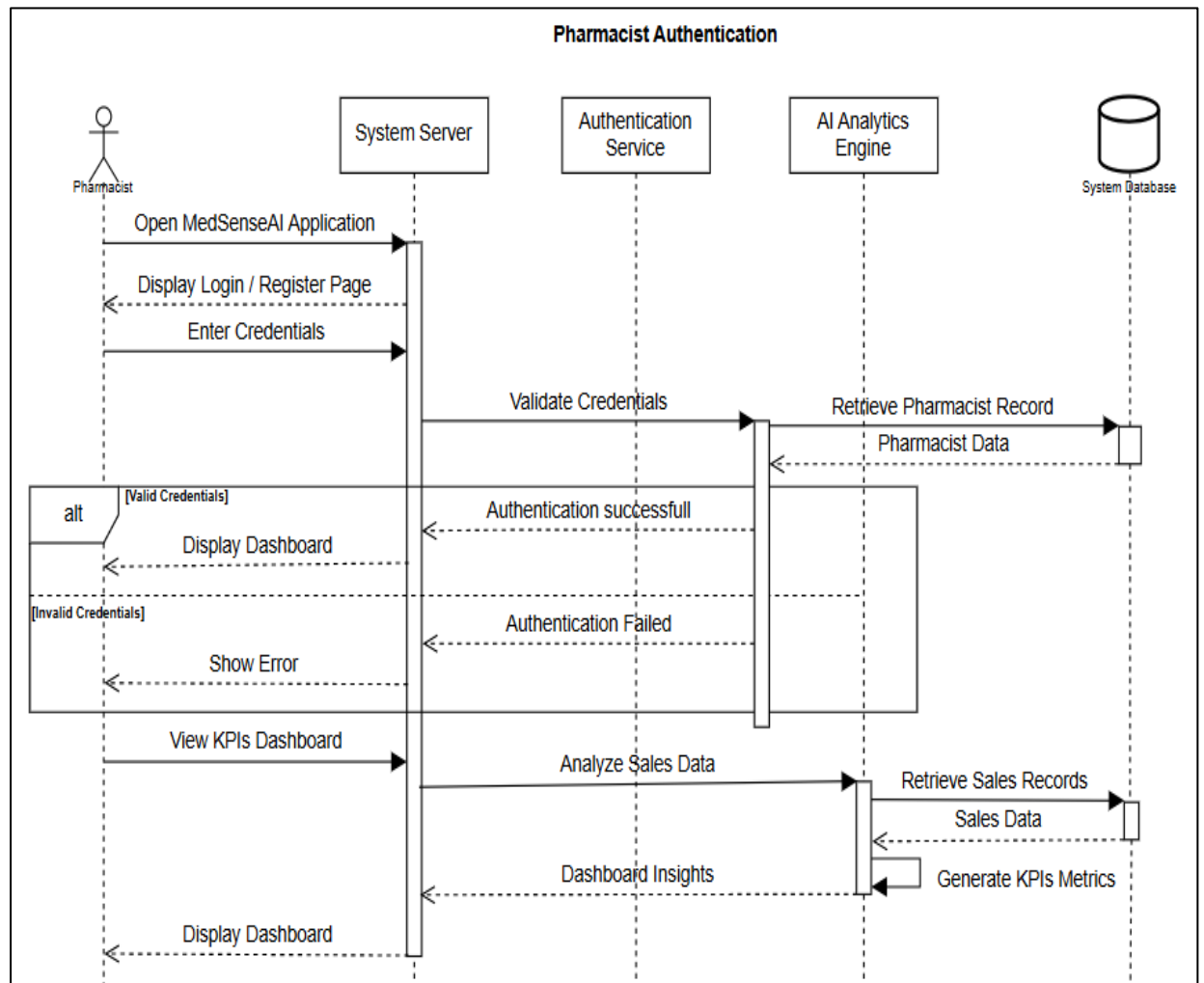


Figure 17: SD-05 Pharmacist Authentication

7.6. Inventory Management

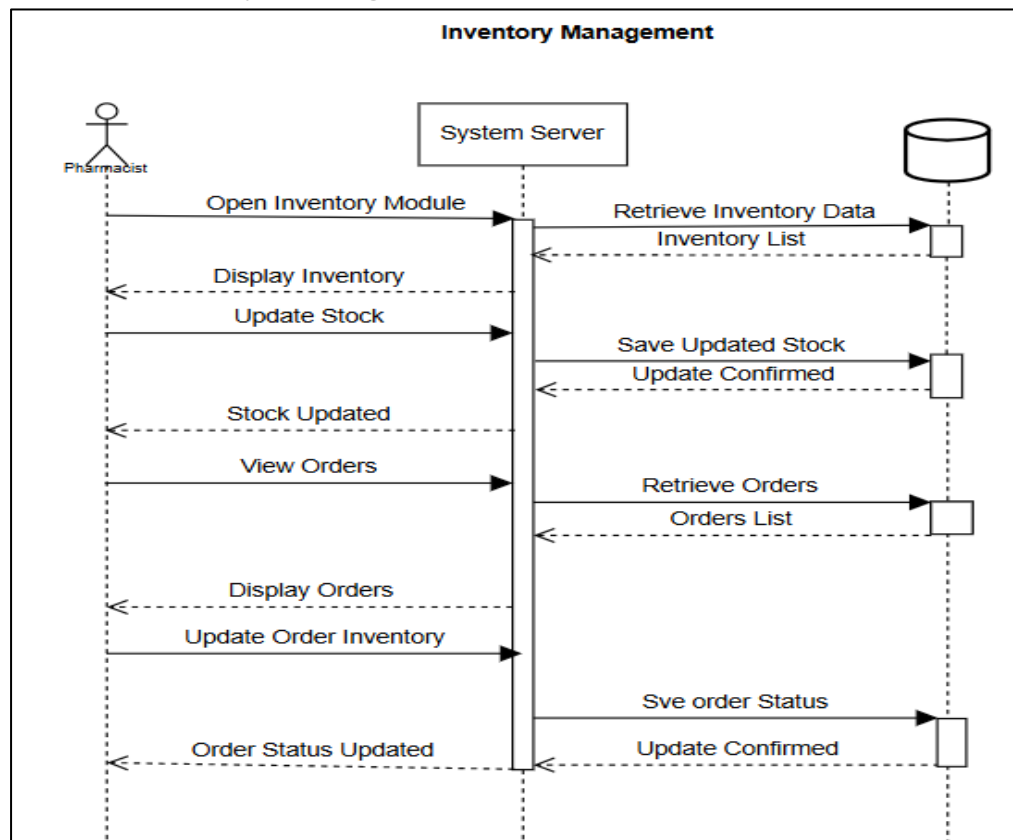


Figure 18: SD-06 Inventory Management

7.7. Dashboard Analytics

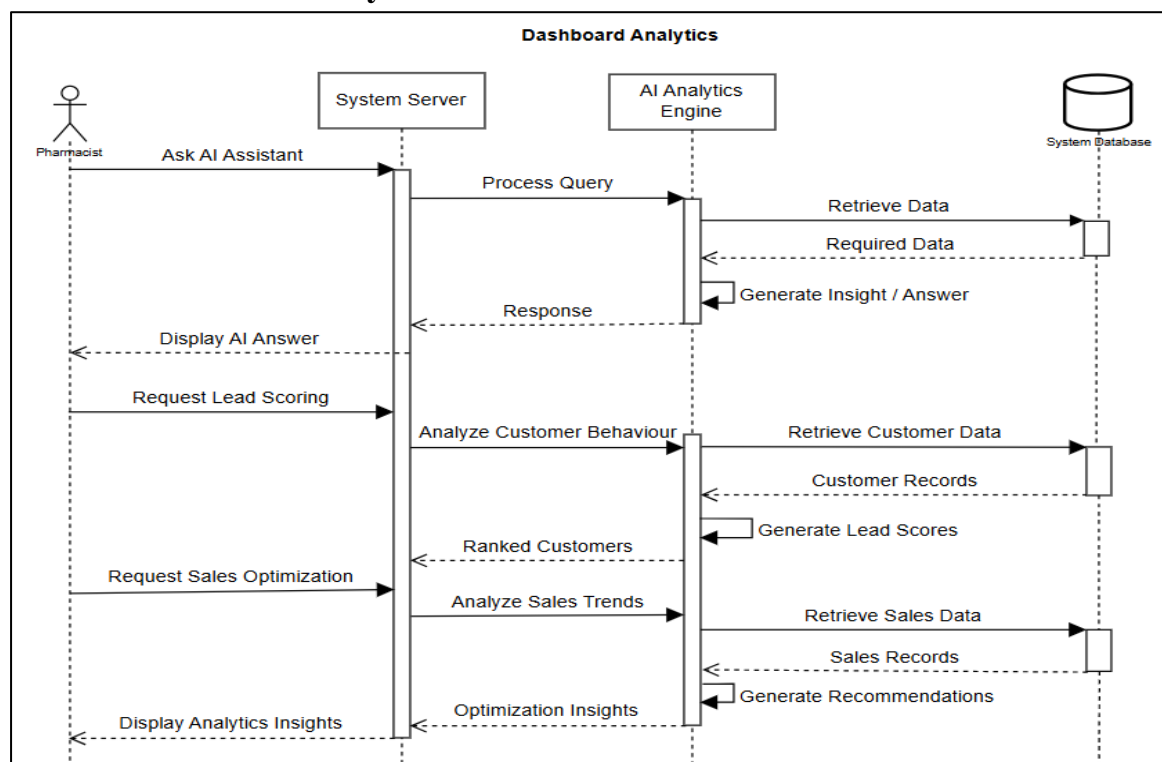


Figure 19: SD-07 Dashboard Analytics

8. Activity Diagram

8.1. User Authentication

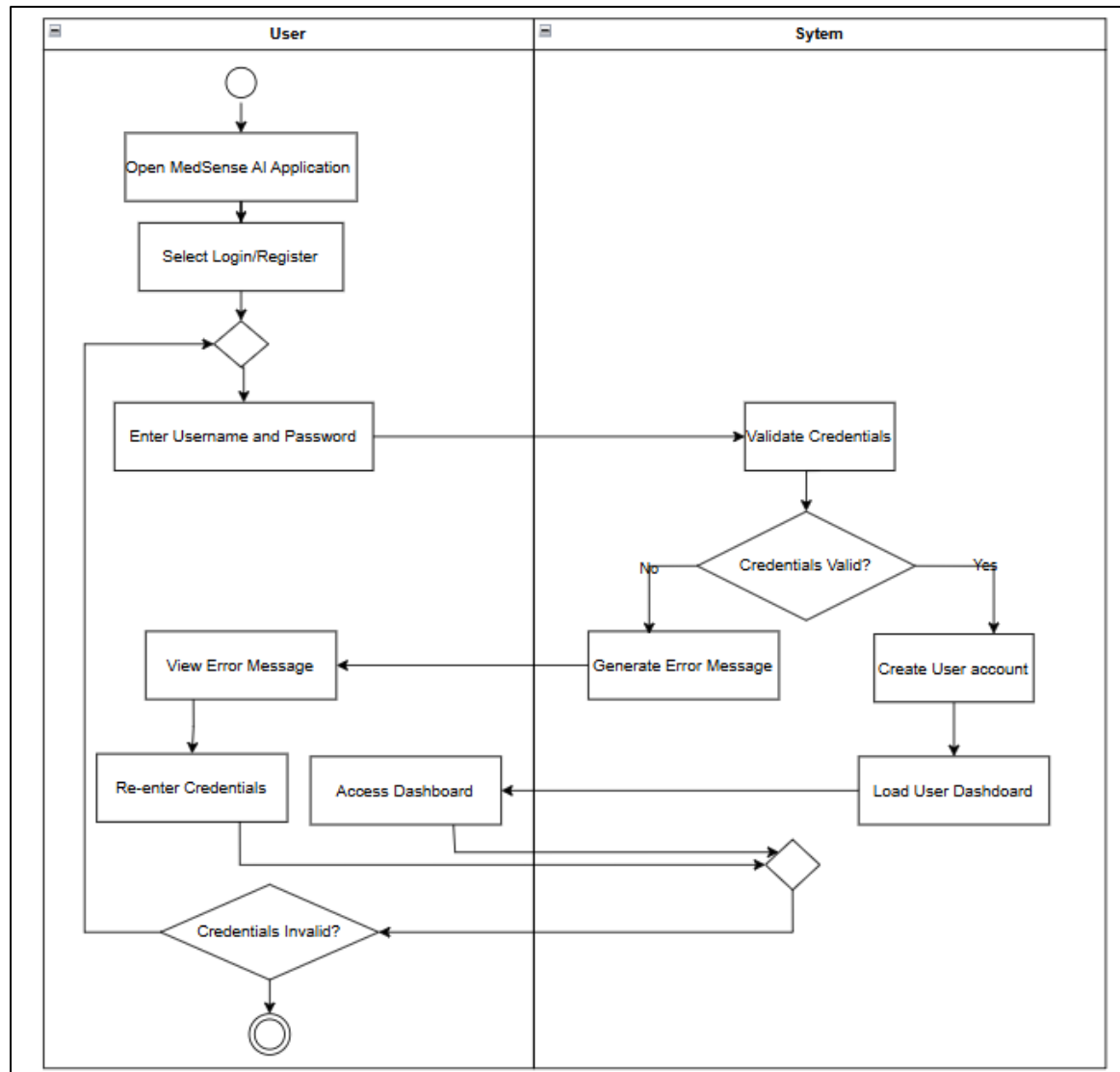


Figure 20: AD-01 User Authentication

8.2. Upload Prescription

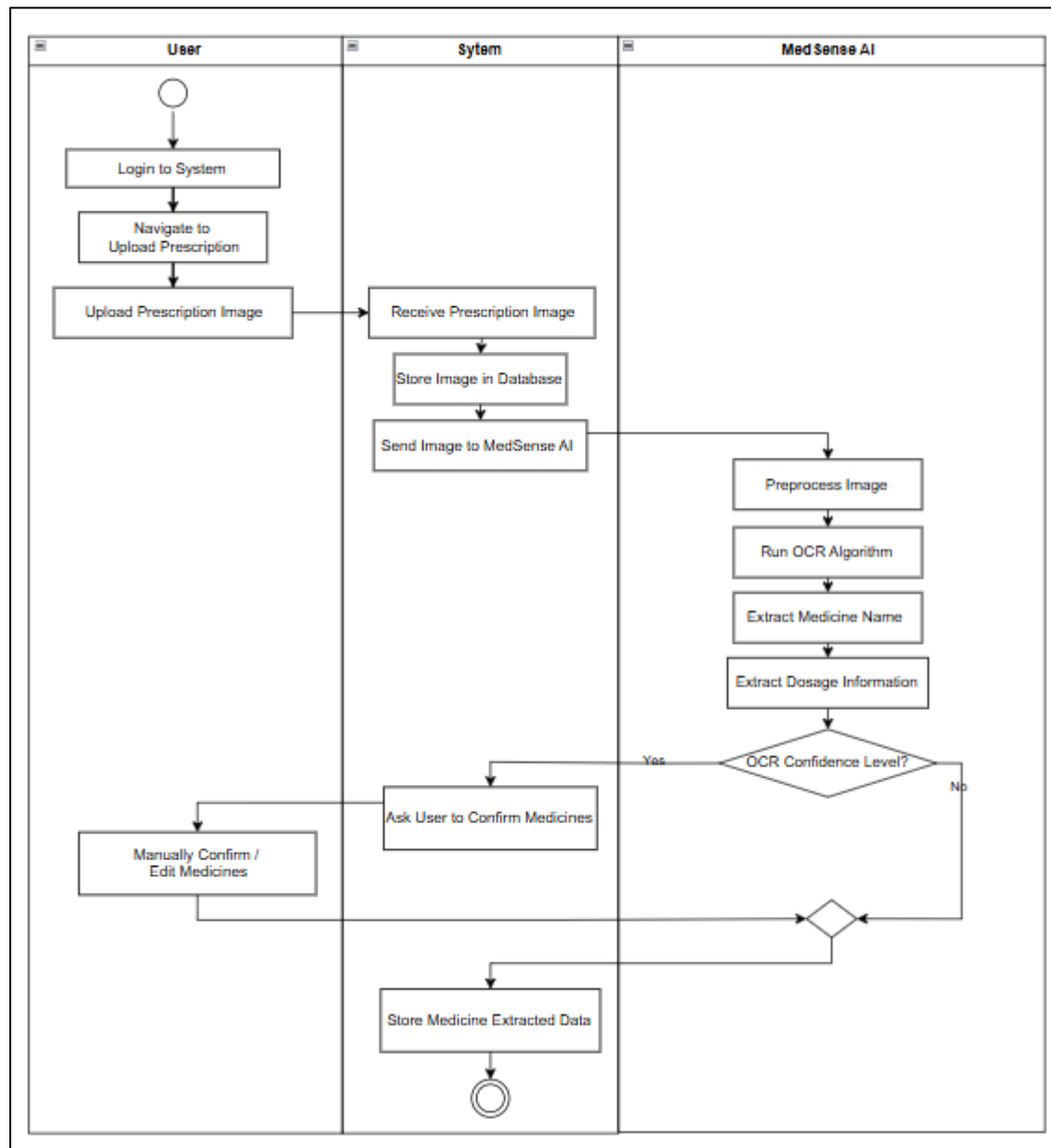


Figure 21: AD-02 Upload Prescription

8.3. Drug Interaction

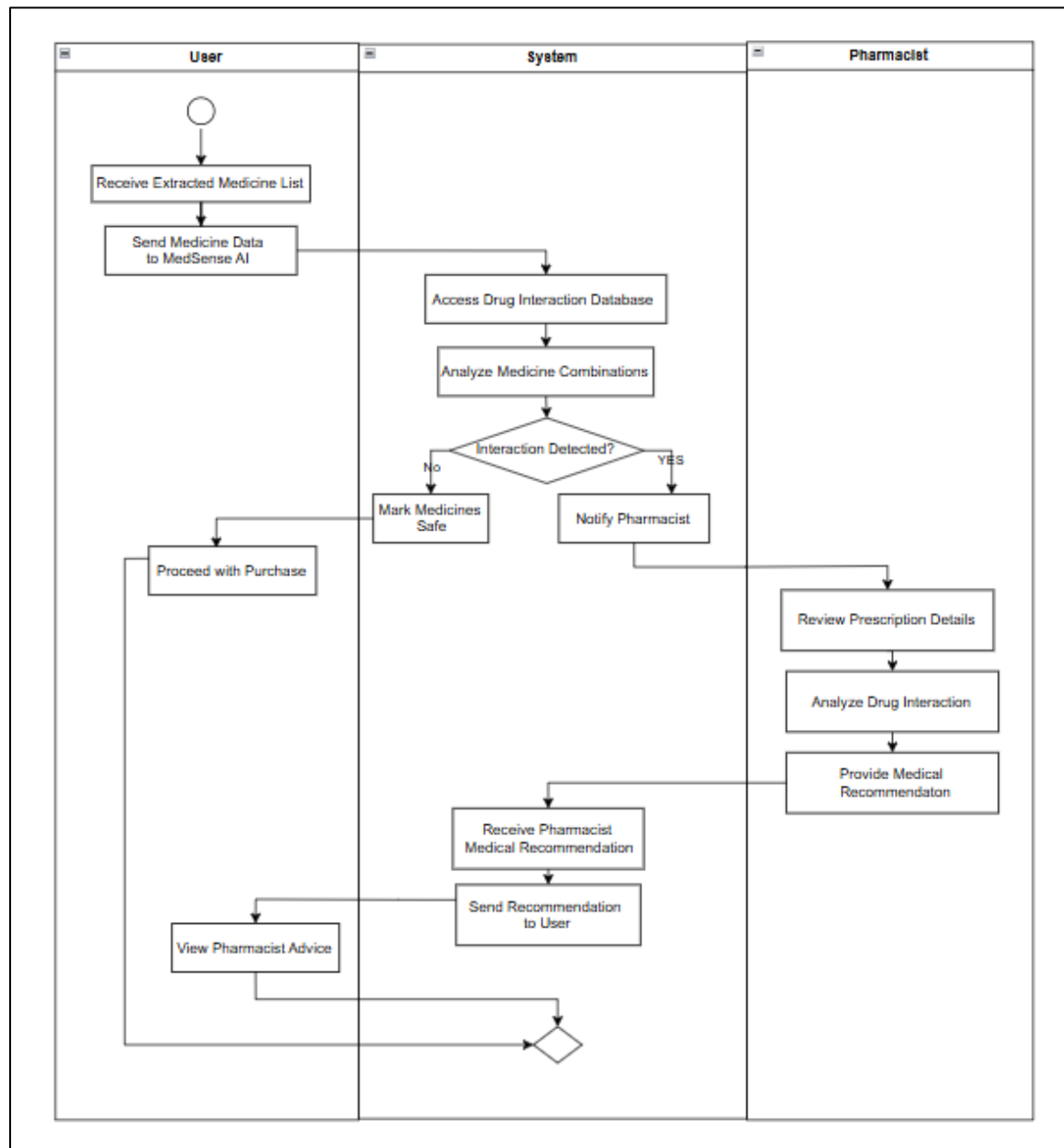


Figure 22: AD-03 Drug Interaction

8.4. Add to Cart

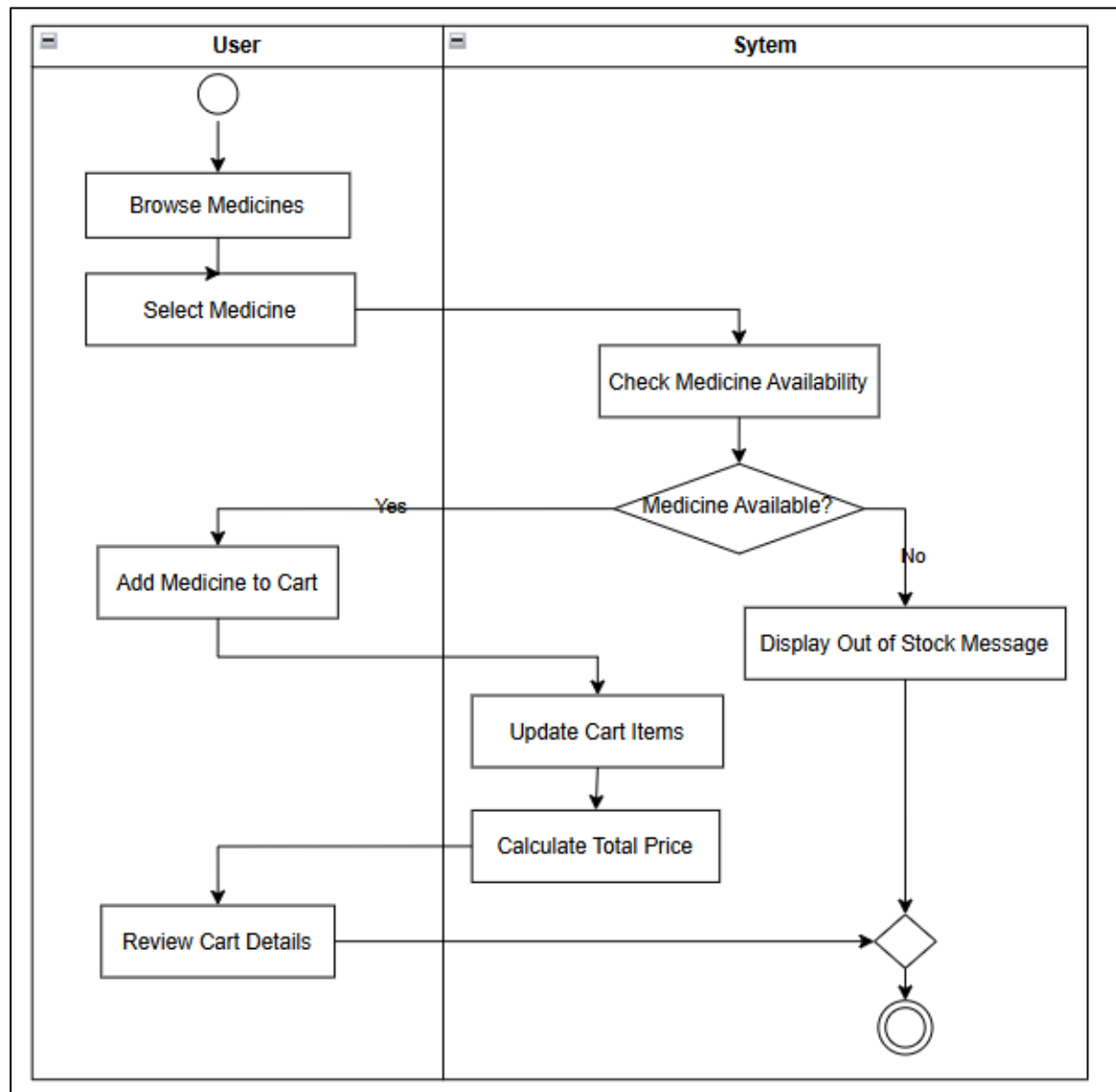


Figure 23: AD-4 Add to Cart

8.5. Refill Notification

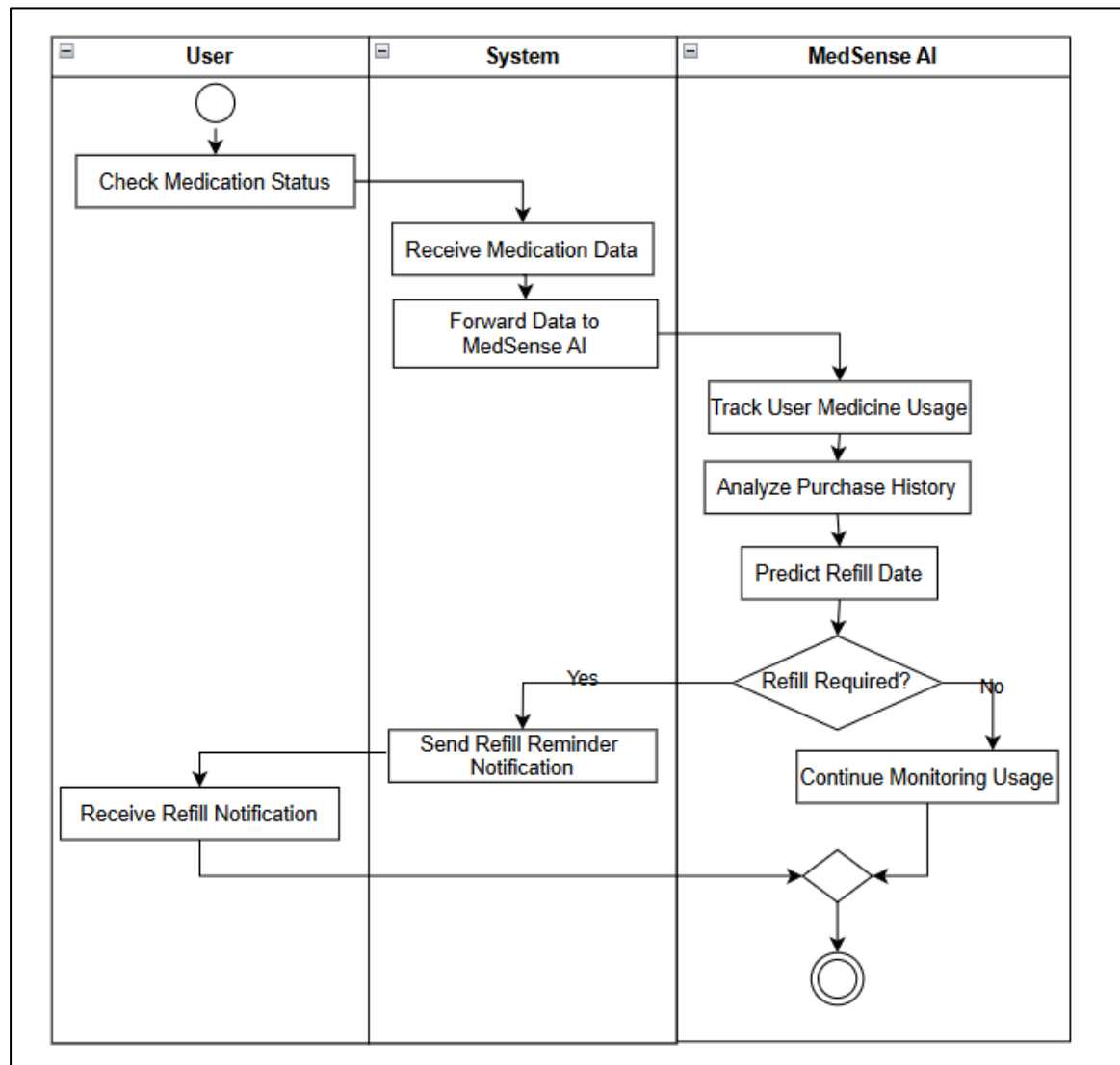


Figure 24: AD-05 Refill Notification

8.6. Pharmacist Consultant

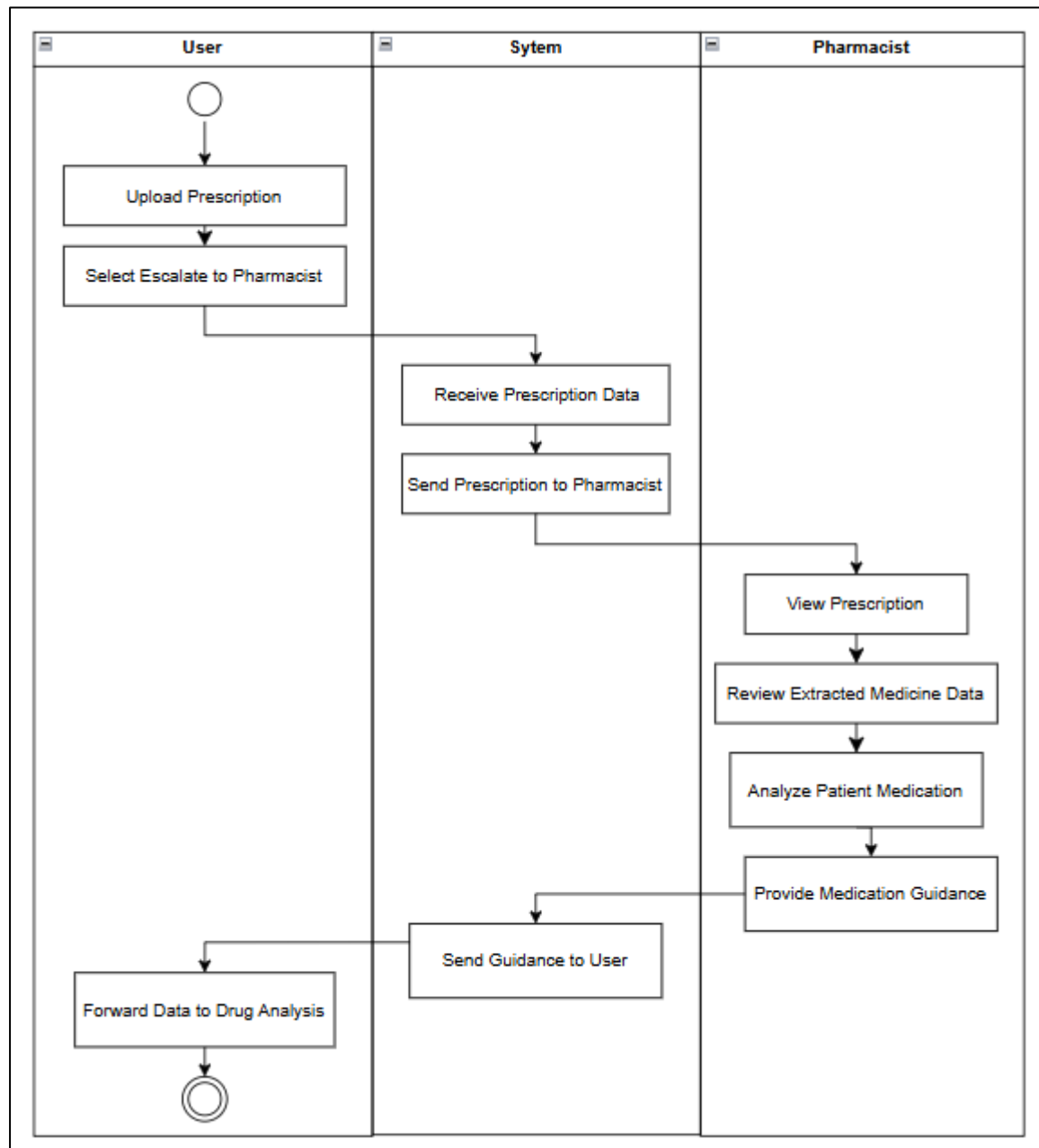


Figure 25: AD-06 Pharmacist Consultant

8.7. Pharmacy Management

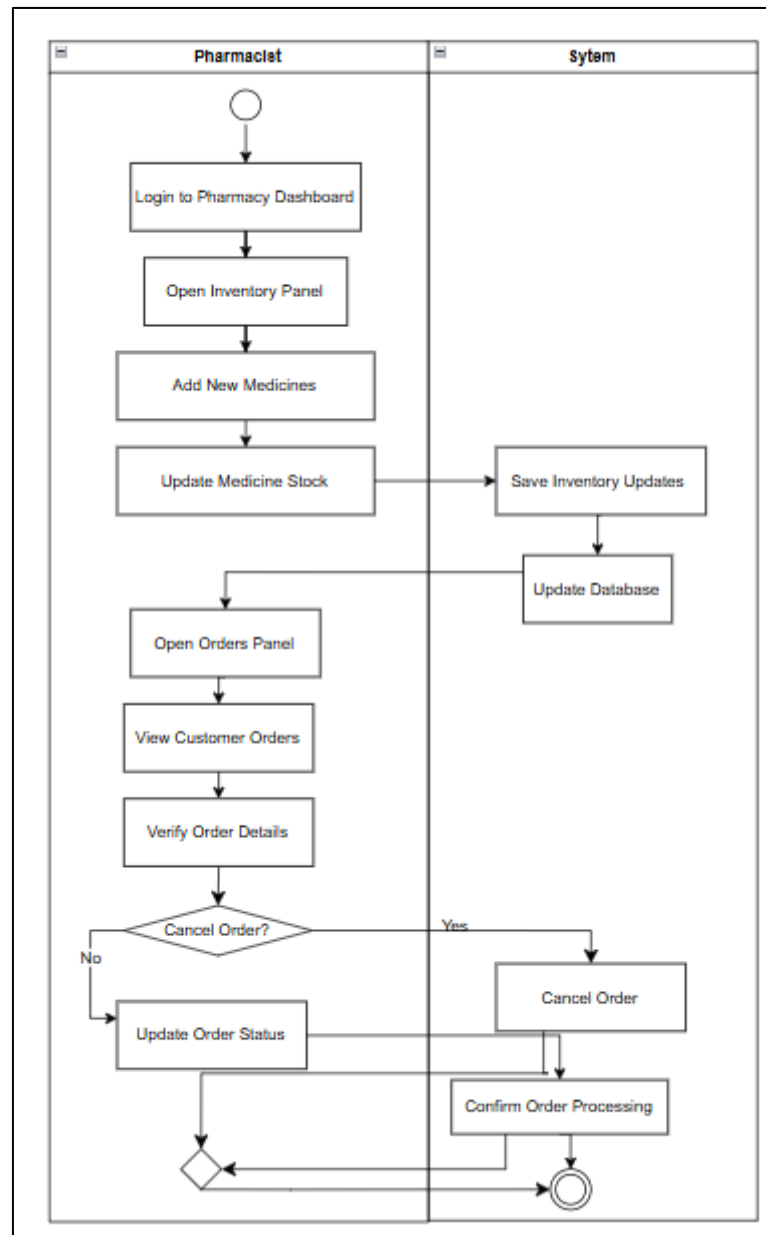


Figure 26: AD-07 Pharmacy Management

8.8. Dashboard Analytics

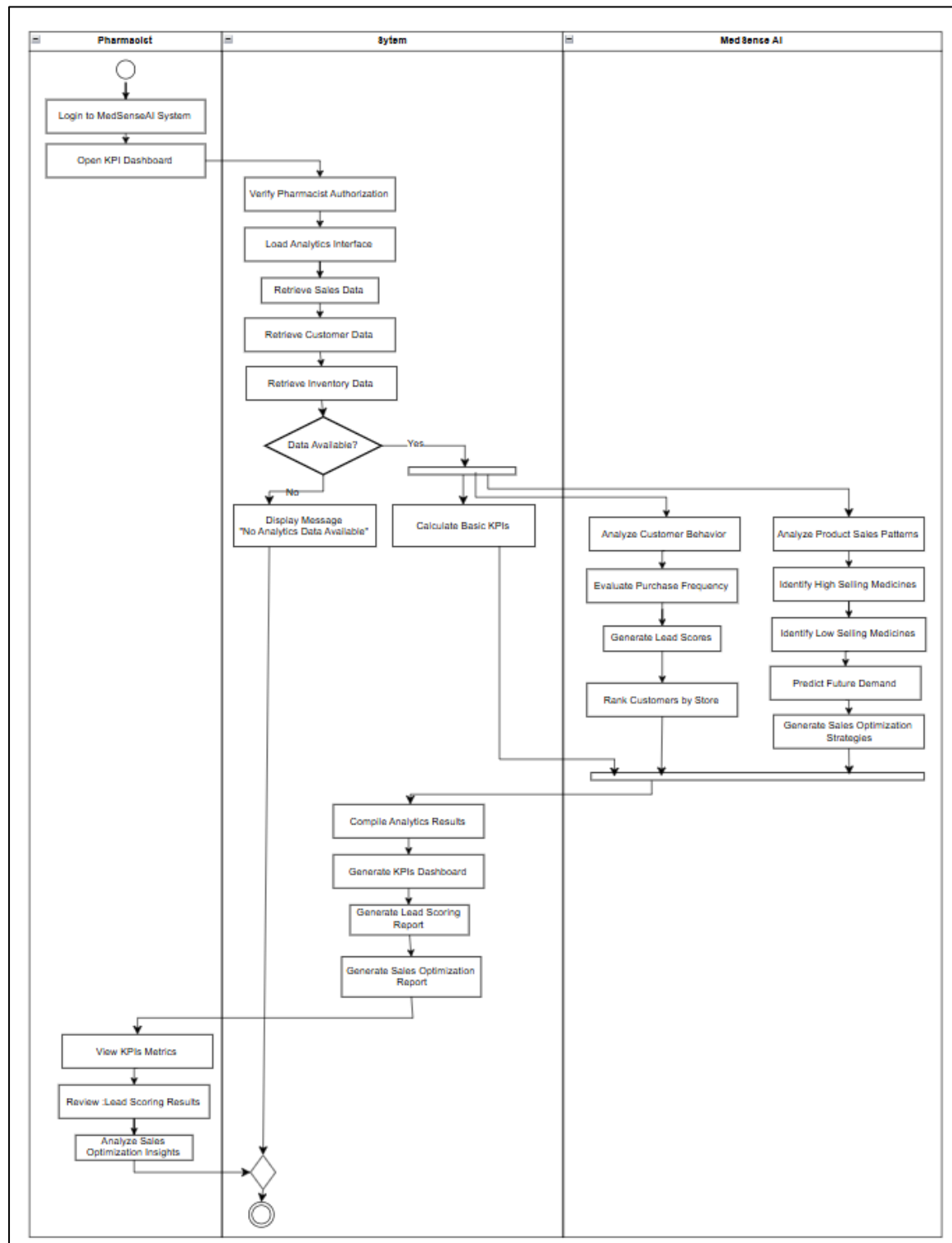


Figure 27: AD-08 Dashboard Analytics

8.9. AI Chatbot

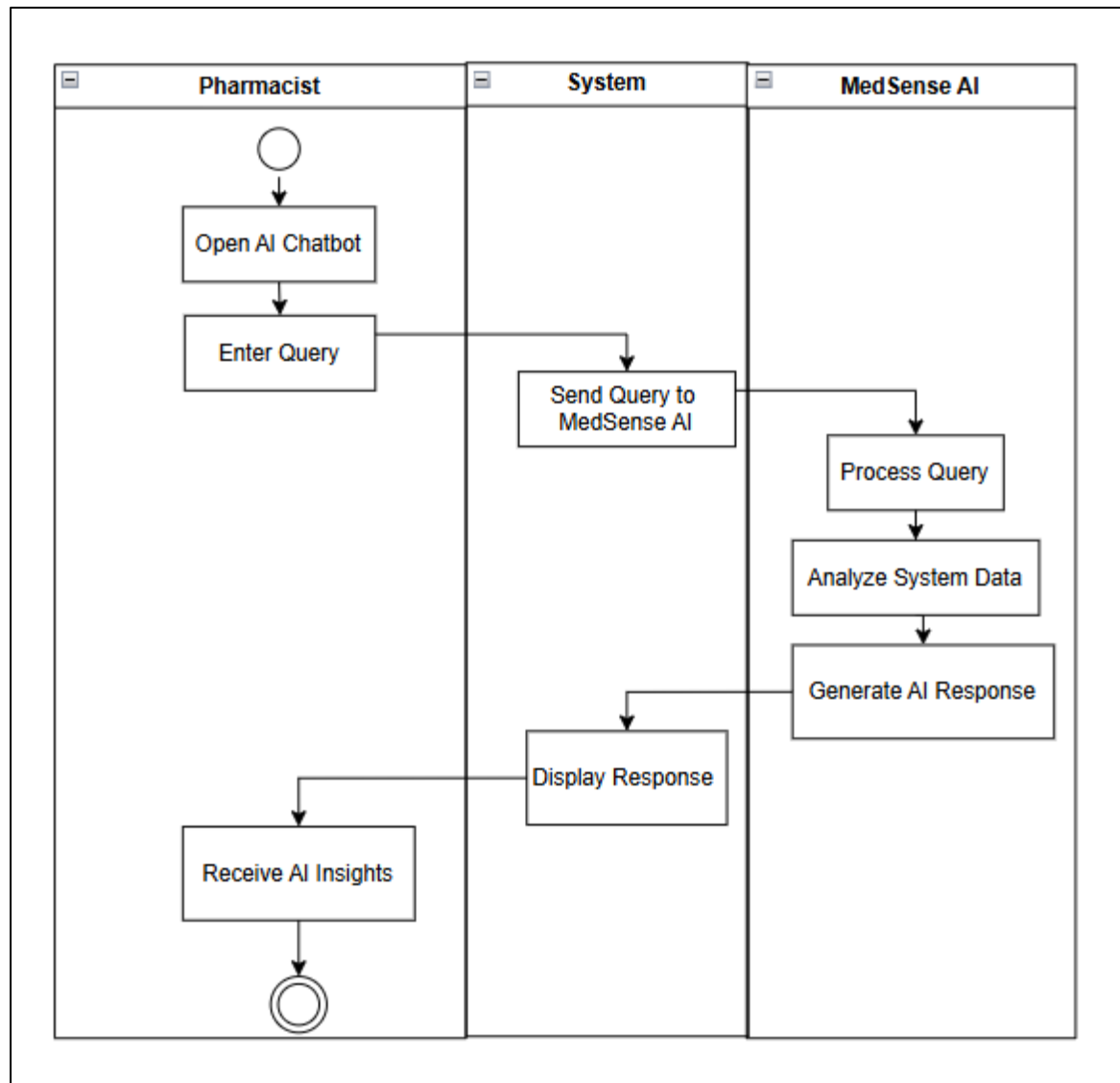
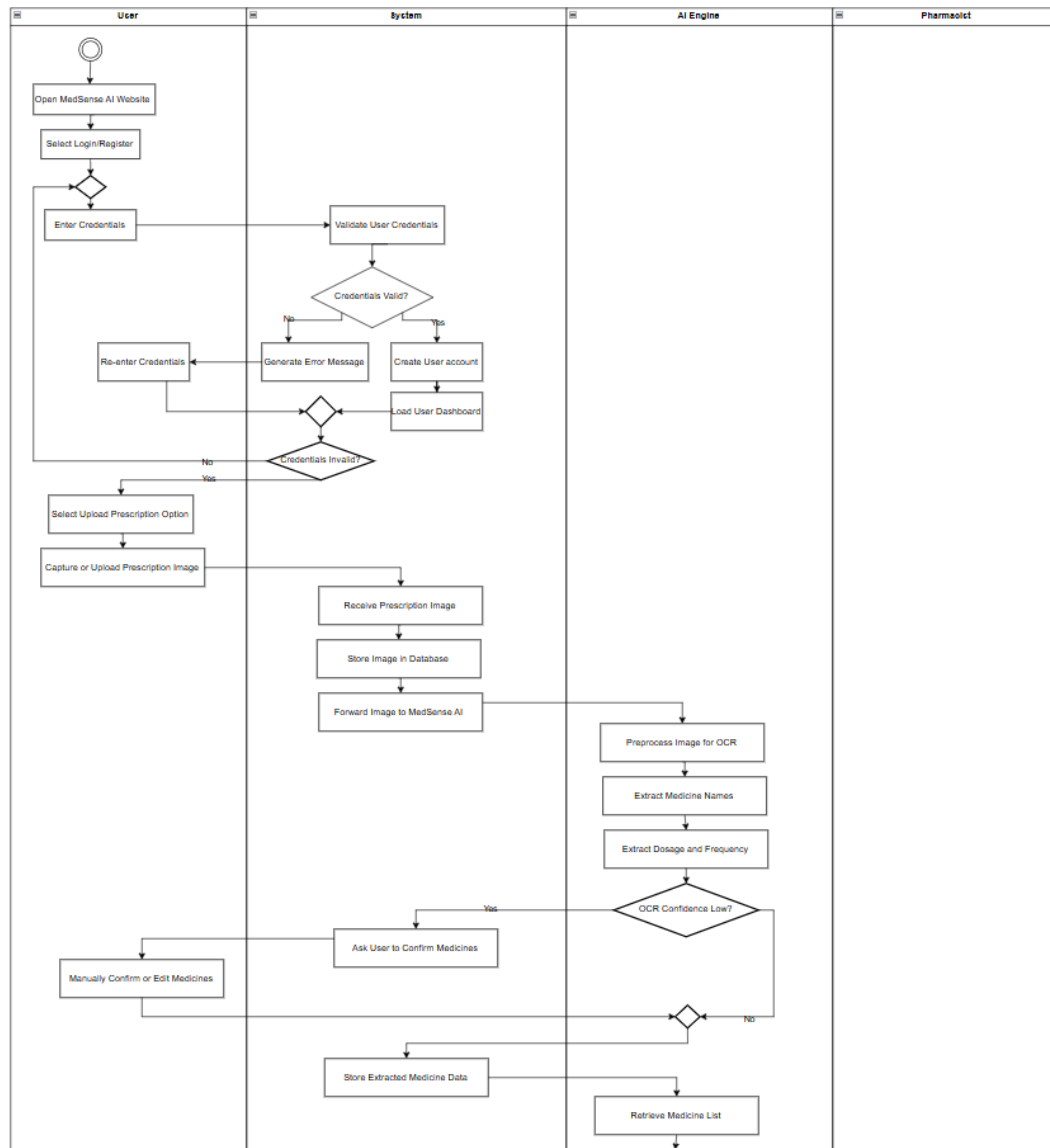
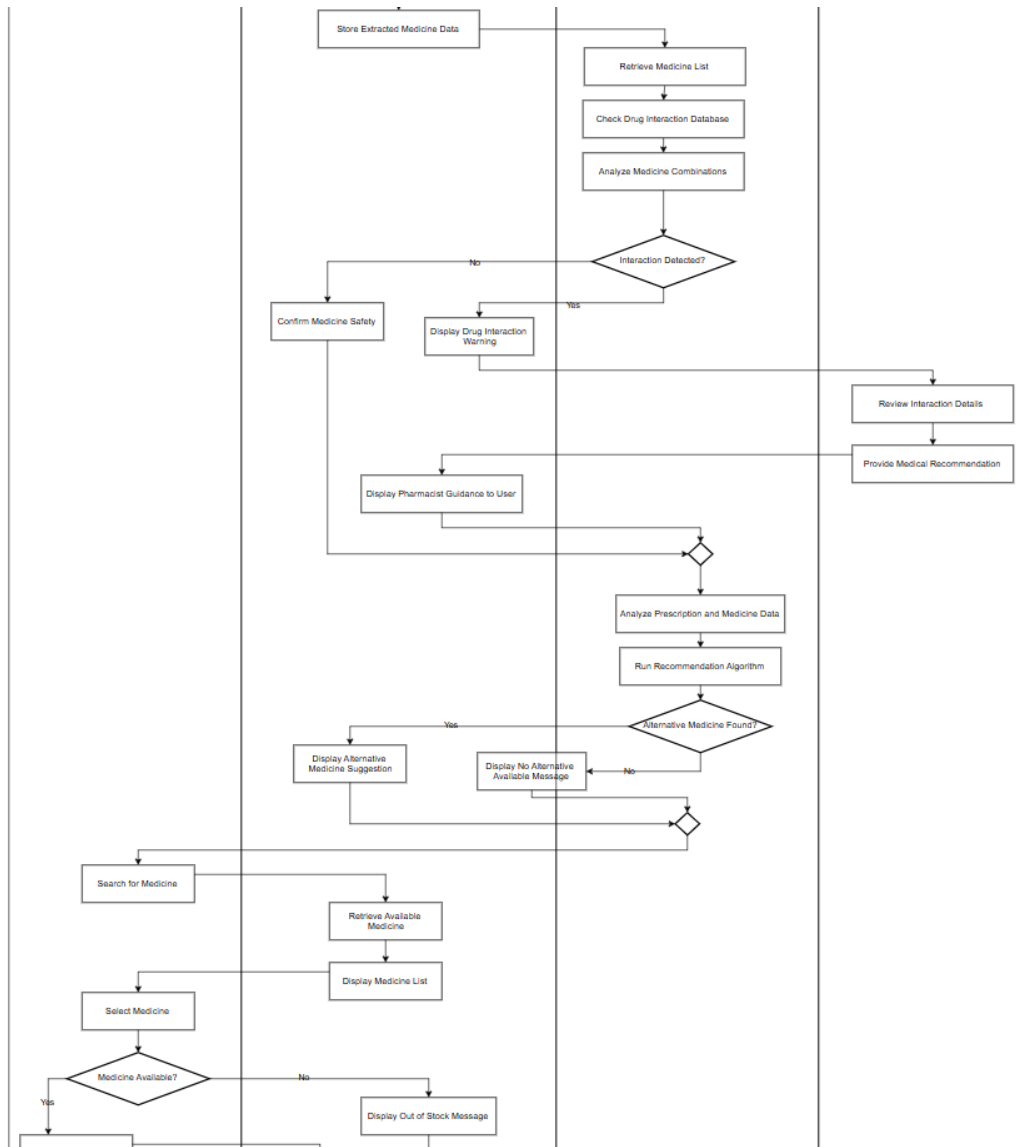
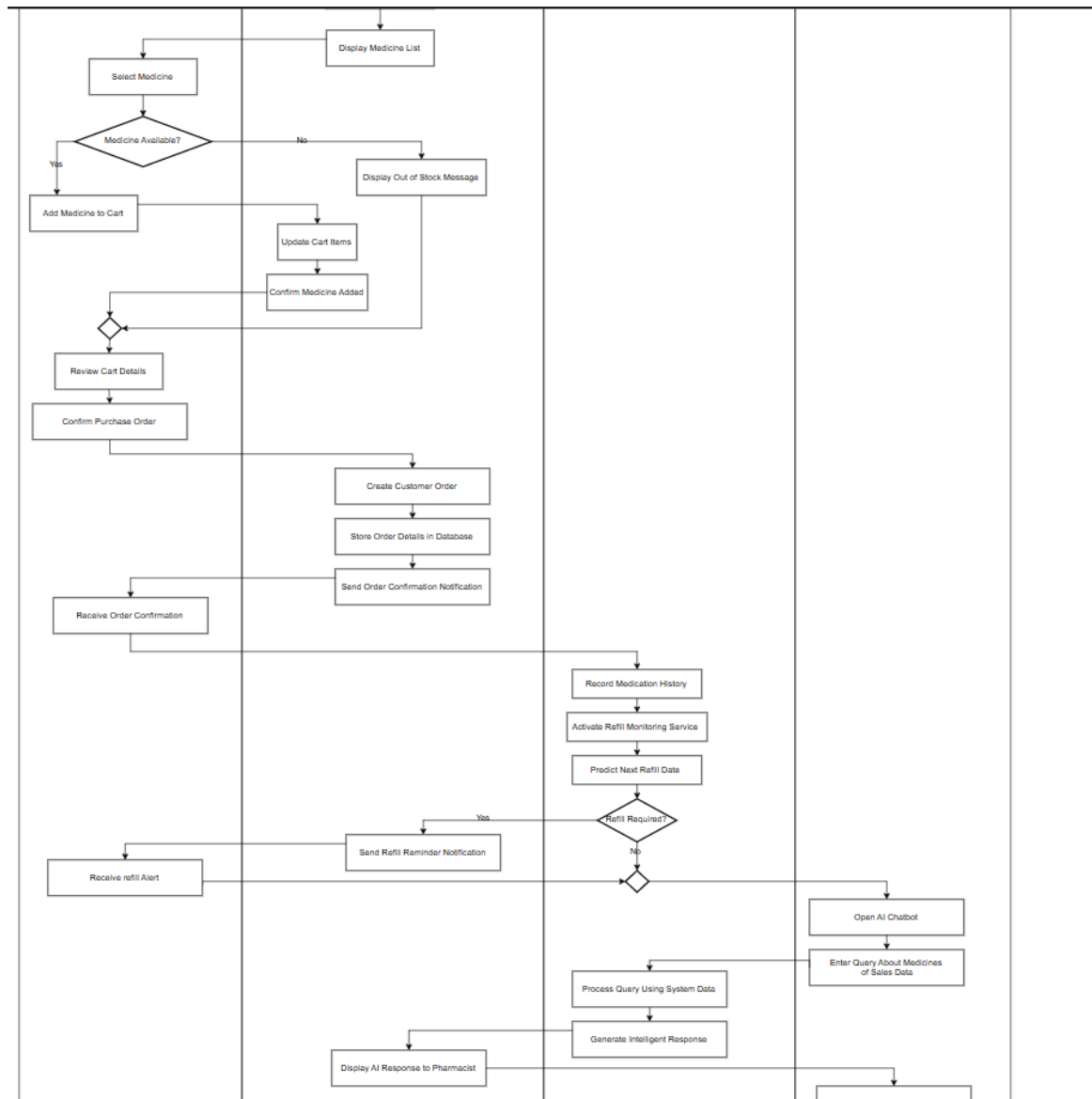


Figure 28: AD-09 AI Chatbot

8.10. Whole System







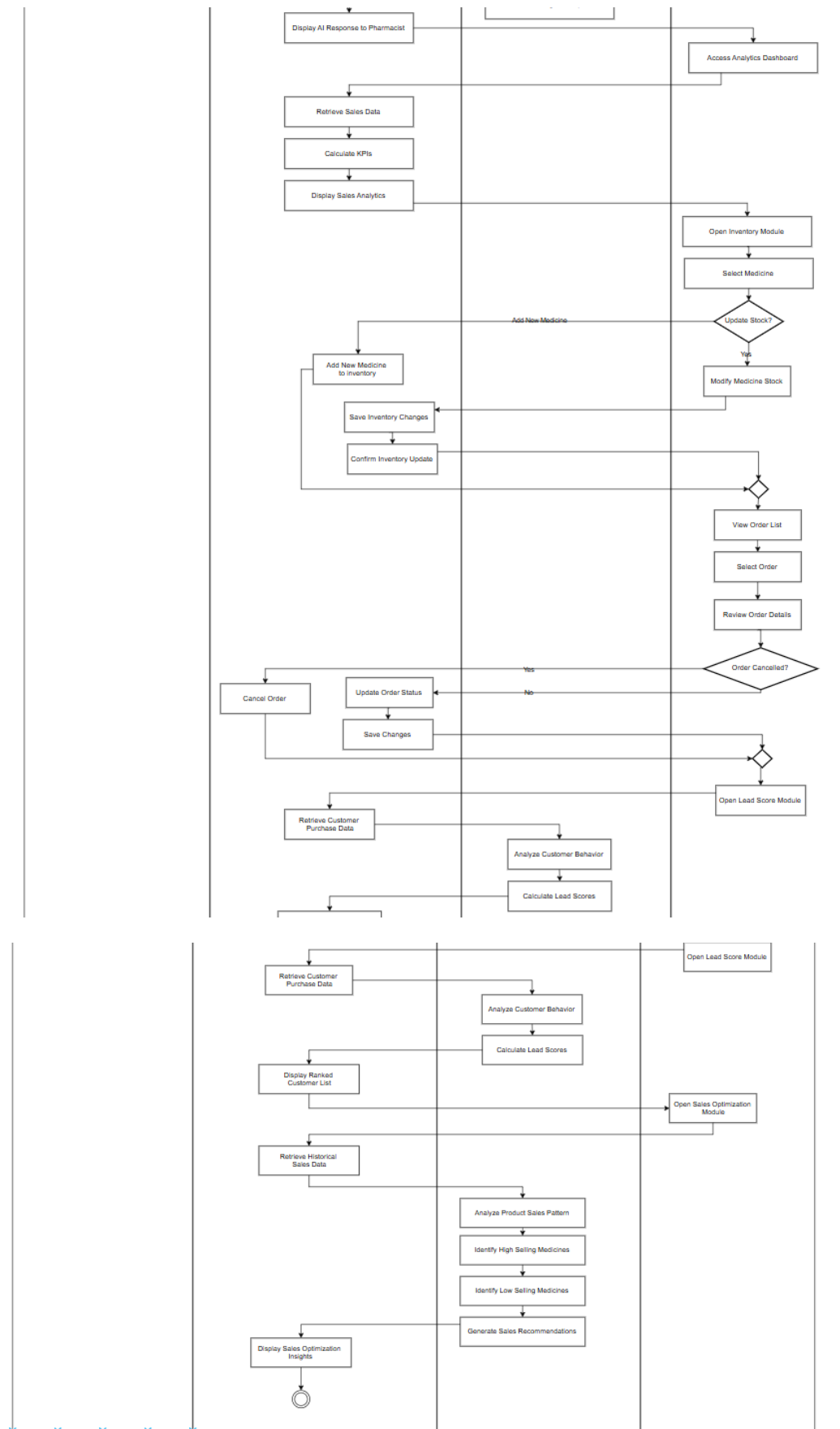


Figure 29: AD-10 WHOLE SYSTEM

9. Class Diagram

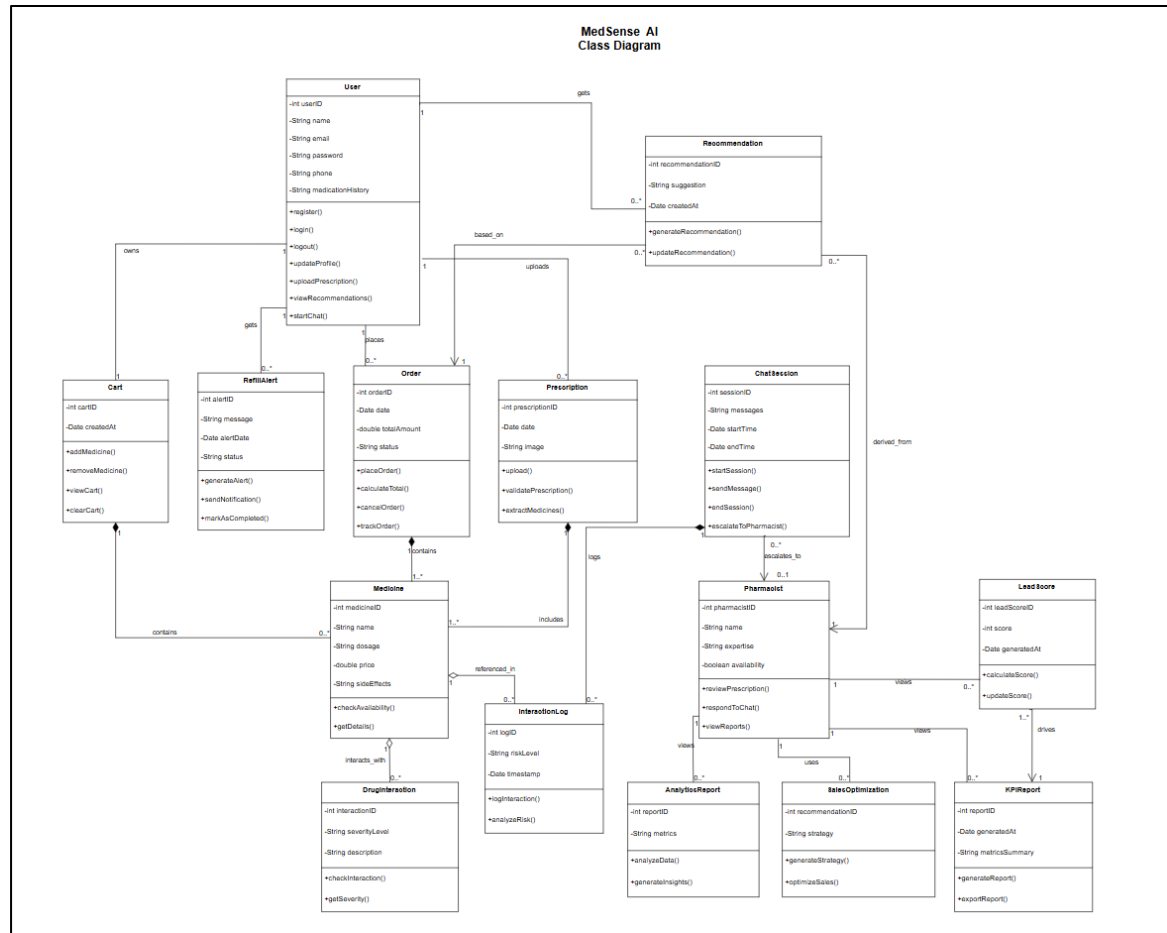


Figure 30: Class Diagram

10. Domain Model

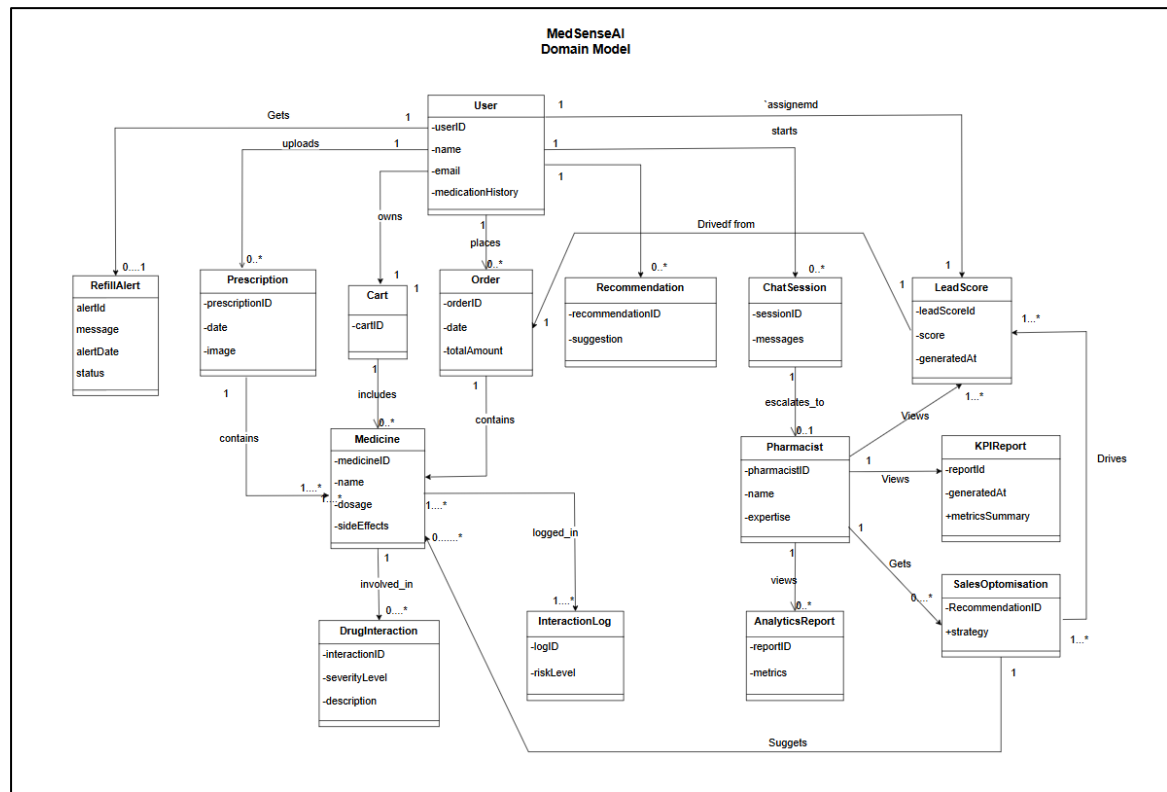


Figure 31: Domain Model

11. Entity Relationship Diagram

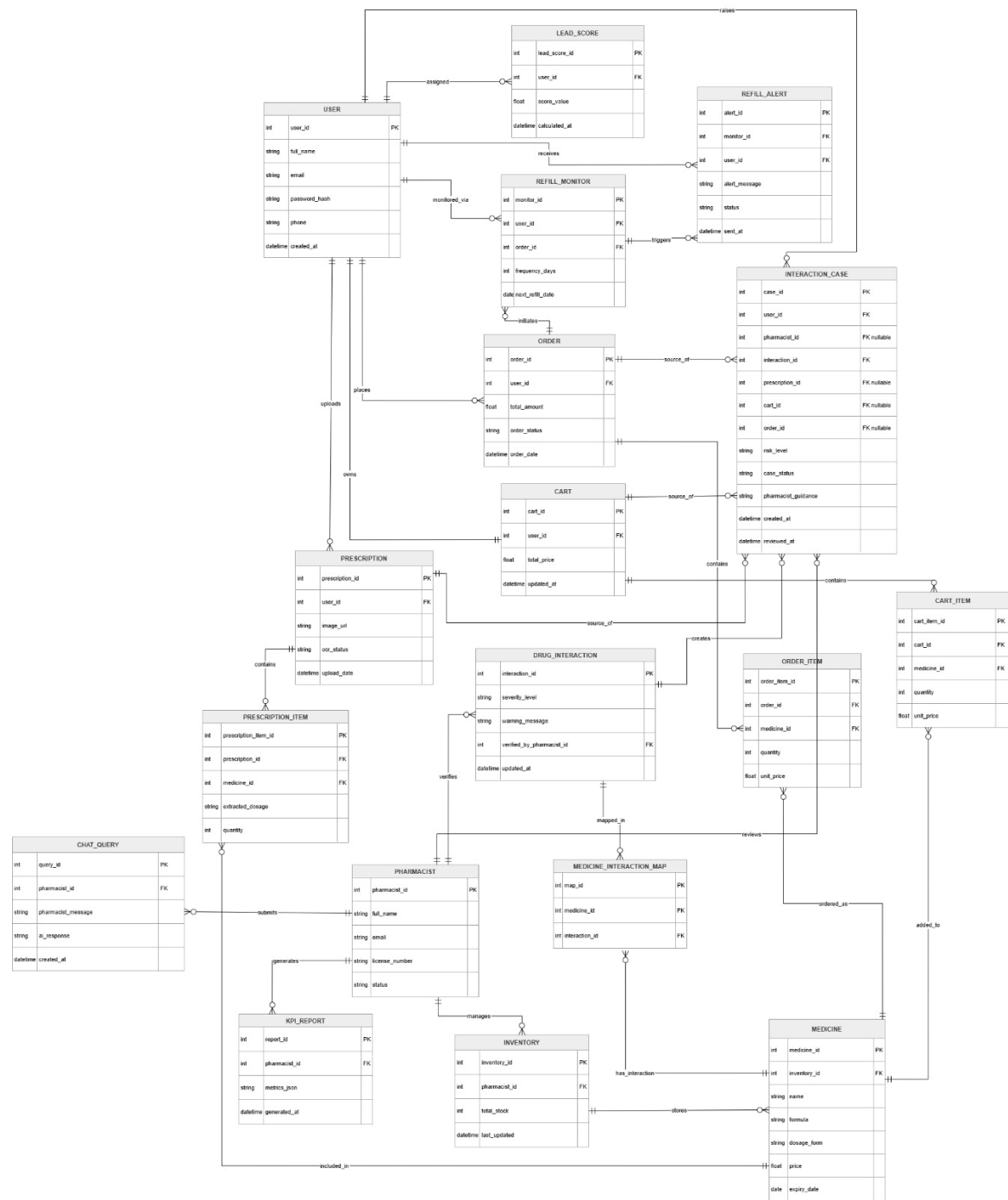


Figure 32: Entity Relationship Diagram

12. Data Flow Diagram

12.1. Level 0

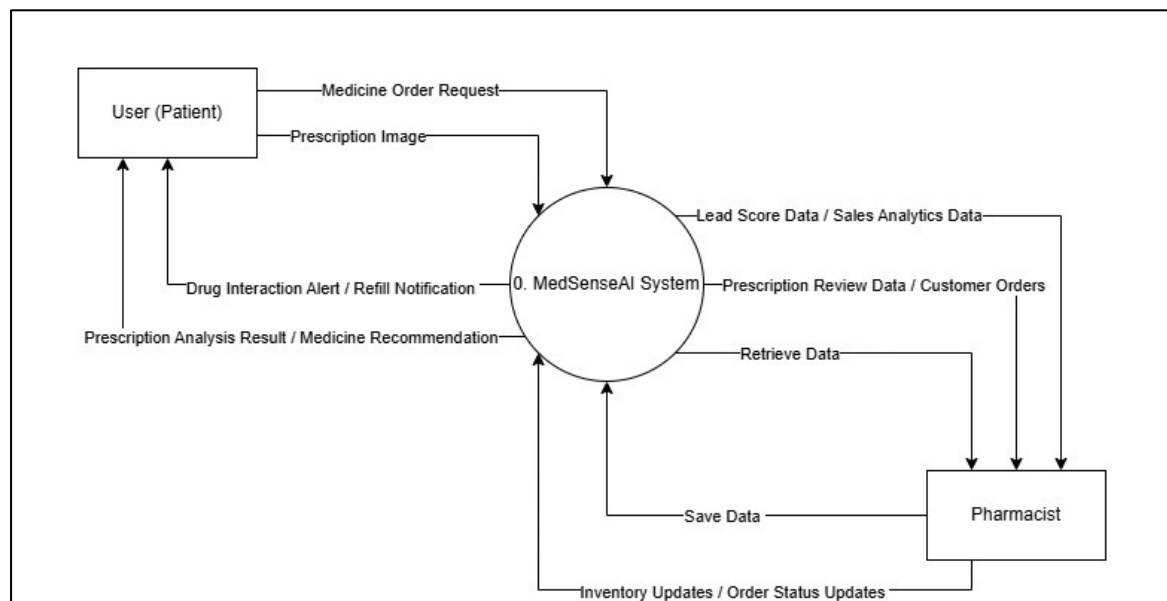


Figure 33: DFD Level-0

12.2. Level 1

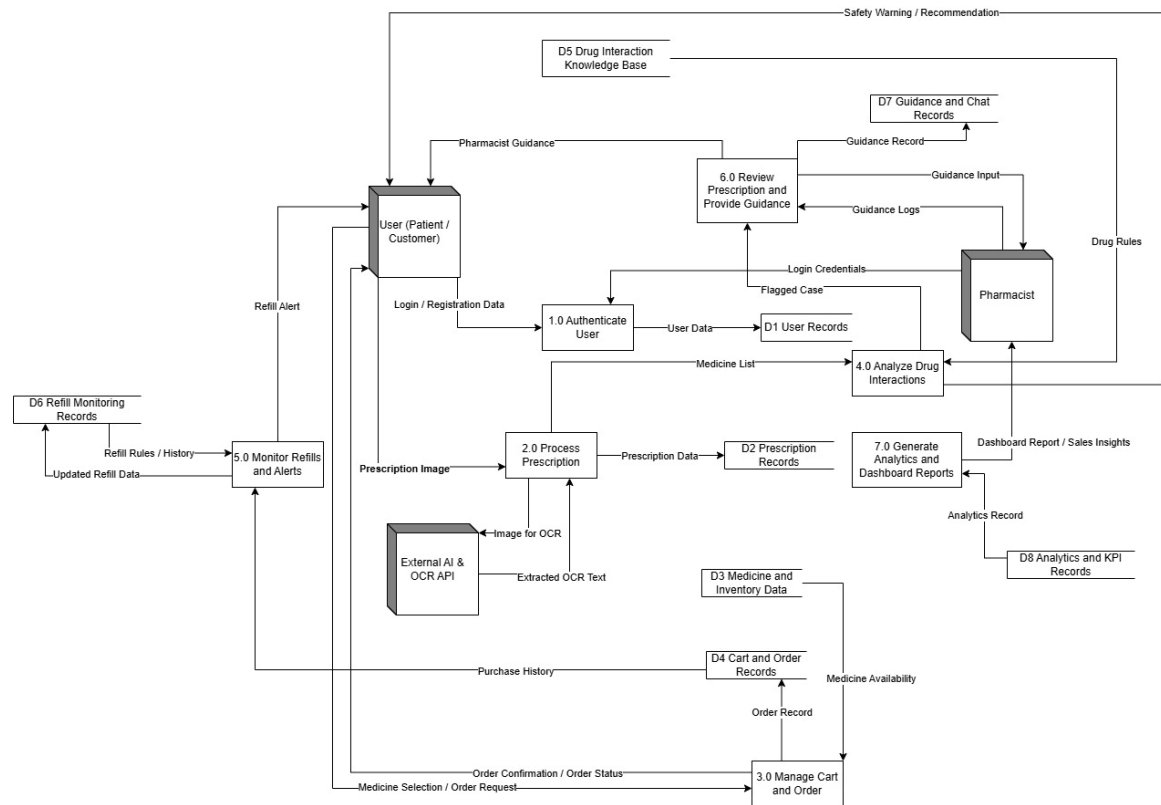


Figure 34: DFD Level-01

12.3. Level 2

12.3.1. Level 2.0

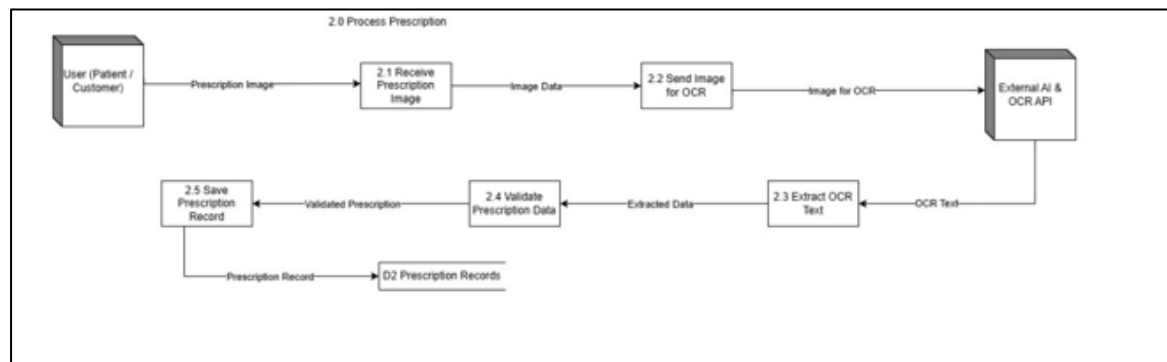


Figure 35: DFD Level-2.0

12.3.2. Level 2.1

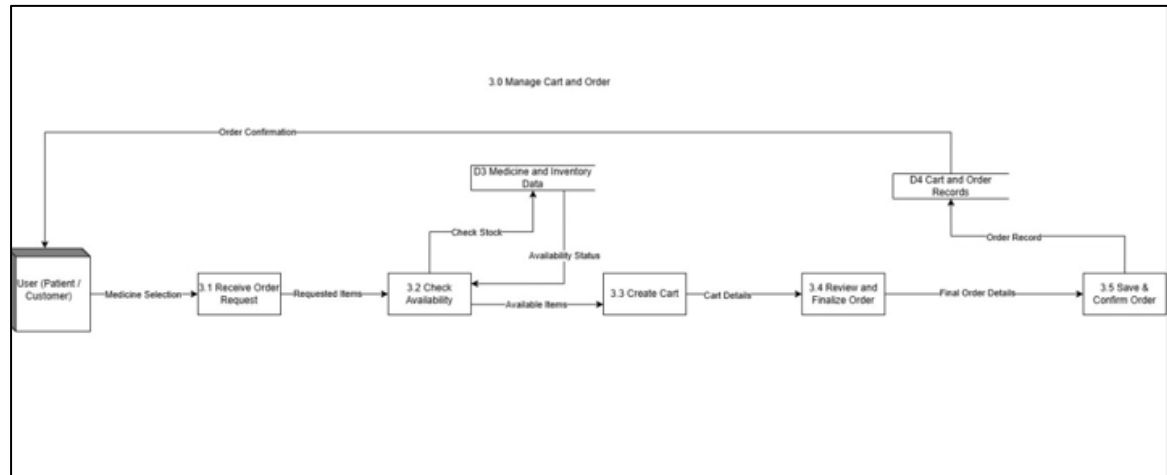


Figure 36: DFD Level-2.1

12.3.3. Level 2.2

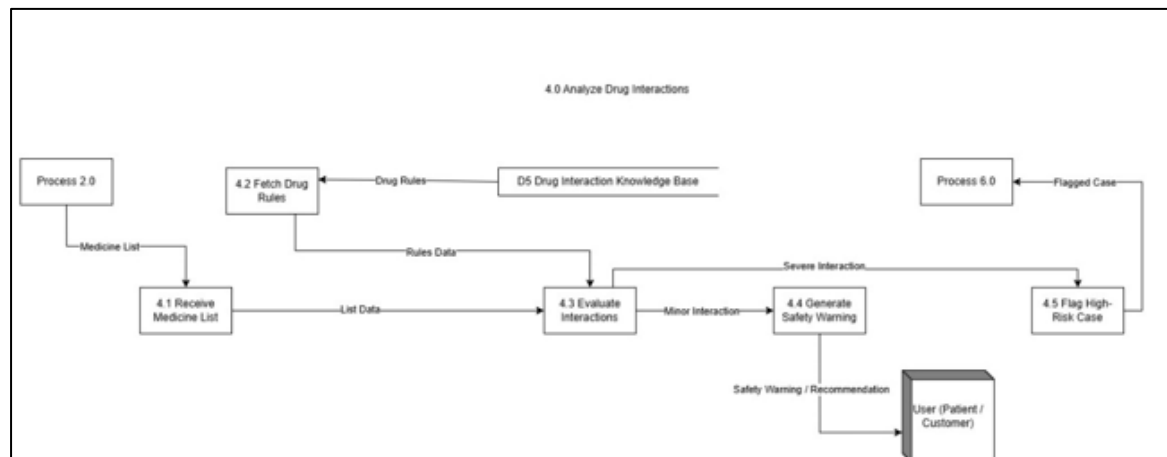


Figure 37: DFD Level-2.2

12.3.4. Level 2.3

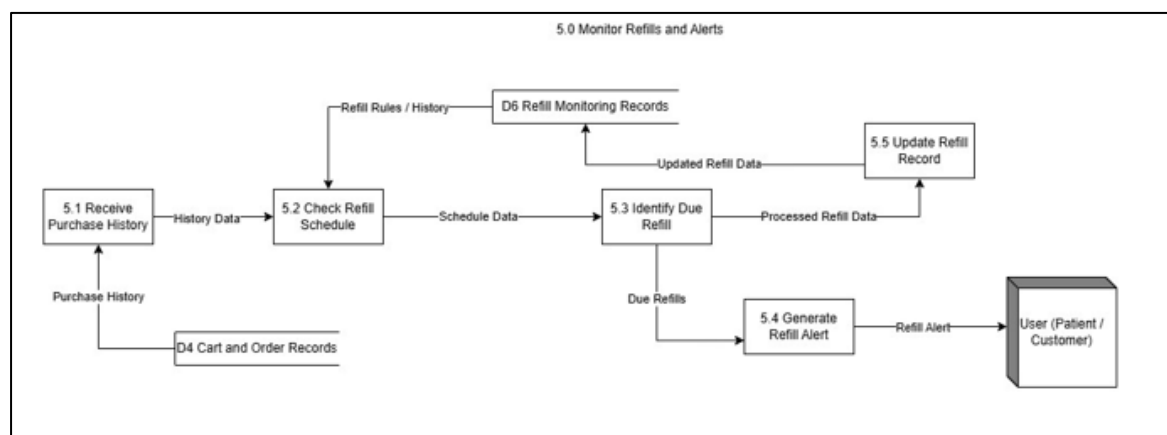


Figure 38: DFD Level-2.3

12.3.5. Level 2.4

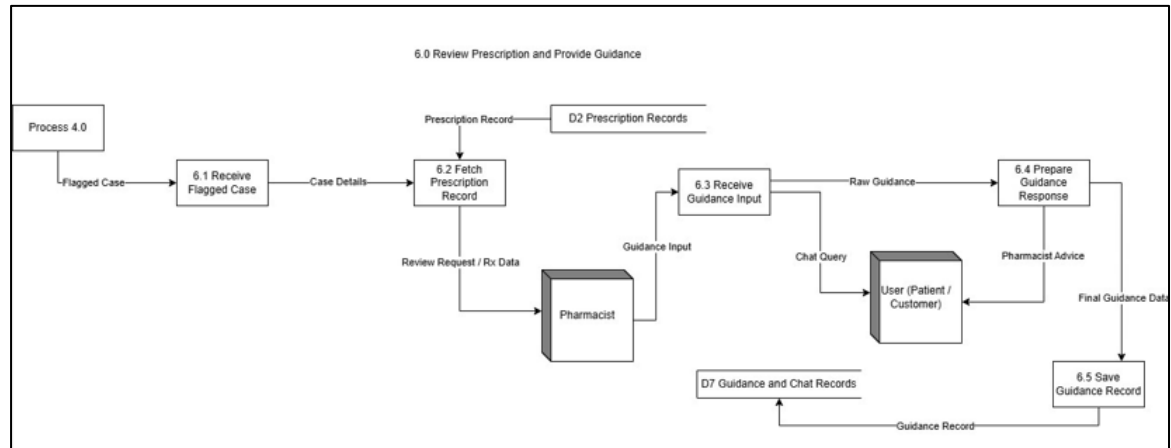


Figure 39: DFD Level-2.4

12.3.6. Level 2.5

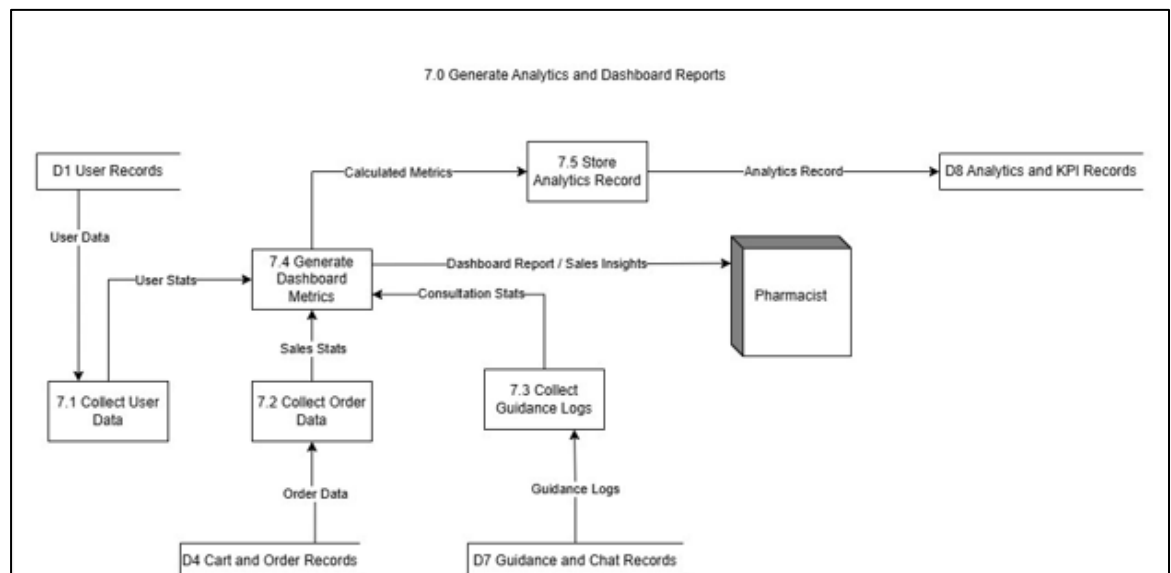


Figure 40: DFD Level-2.5

13. References

- [1] <https://ieeexplore.ieee.org/document/10968850/>
- [2] <https://www.medsense.health/>
- [3] <https://www.comfygen.com/blog/ai-pharmacy-app-development/>